

Cybersecurity & Cyber-Resilience Strategy for the Sovereign Cloud–Edge–IoT Continuum

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Abstract

The rapid decentralization of computing infrastructures toward the Cloud–Edge–IoT continuum introduces unprecedented cybersecurity challenges, including expanded attack surfaces, heterogeneous environments, and increased dependency on non-sovereign technologies. This paper proposes a structured, multi-stage methodology for defining and prioritizing cybersecurity capabilities required to achieve a secure and sovereign European computing continuum. The approach combines a pipeline-oriented transformation model with a seven-dimensional analytical framework, enabling traceable transitions from architectural capabilities to strategic priorities and policy-aligned recommendations. A four-filter prioritization logic identifies critical capabilities based on regulatory urgency, architectural dependencies, sovereignty risks, and market feasibility. These priorities are further analyzed through a progression matrix capturing operational deployment and strategic sovereignty evolution across time phases. The methodology provides a reproducible and evidence-based framework for guiding research, innovation investments, and policy decisions in the context of the European digital sovereignty agenda.

Keywords: *Cybersecurity, Cloud-Edge-IoT, Digital Sovereignty, Zero Trust, Post-Quantum Cryptography, EU Policy, Cyber-Resilience*

1 Introduction

1.1 Contributions

This paper makes the following contributions:

The methodology addresses a gap identified across three mature but previously separate research traditions: technology roadmapping (TRM), capability-based planning (CBP), and cybersecurity capability maturity assessment. To the best of our knowledge, no existing framework integrates all four elements—(a) architecture-anchored capability derivation, (b) multi-criteria prioritisation, (c) multi-dimensional profiling across technical, regulatory, market, and sovereignty axes, and (d) phased progression with quantitative maturity metrics—in a unified, traceable pipeline tailored to the Cloud–Edge–IoT cybersecurity domain.

Canonical TRM frameworks [1, 2] provide the phased planning structure but do not natively encode regulatory, sovereignty, or market dimensions. Capability-based planning [3] provides the outcome-oriented capability concept and DOTMLPF-P decomposition logic but has not been systematically applied to civilian cybersecurity roadmapping. EU cybersecurity roadmap precedents—CyberROAD [4], CONCORDIA [5], CyberSec4Europe [6]—provide multi-dimensional framing and phased recommendations but lack quantitative per-capability maturity metrics and architecture-anchored capability derivation. The present methodology synthesises these traditions into a single reproducible instrument.

- A structured pipeline methodology for cybersecurity capability prioritization in the Cloud–Edge–IoT continuum.
- A 4-dimensional filtering logic linking regulatory, architectural, sovereignty, and market-adoption considerations.
- A hybrid 7-dimensional profiling framework for actionable R&I prioritization.
- A progression matrix model capturing operational deployment and strategic sovereignty evolution across time phases.
- A traceable transformation from technical capabilities to policy-aligned strategic recommendations.

This manuscript and roadmap defines the architectural guardrails required to secure the Cloud–Edge–IoT Continuum. It moves beyond traditional “perimeter security” toward a model of cyber-resilience—where systems are engineered to proactively anticipate, withstand, recover from, and adapt to attacks while maintaining strict digital sovereignty. **Approach:** To provide actionable, risk-aware Research & Innovation (R&I) priorities, the strategy is organized into a two-part framework:

Purpose of This Document

This roadmap provides a structured methodology for European policymakers, research organizations, and industry stakeholders to:

- Identify and prioritize cybersecurity R&I investments for the Cloud-Edge-IoT continuum
- Track capability maturity using objective Operational Deployment Phase (ODP) and Strategic Sovereignty Level (SSL) metrics.
- Align Horizon Europe funding with strategic autonomy goals and regulatory compliance requirements
- Ensure coordinated implementation across Member States and European initiatives (ECCC, Digital Europe Programme, EU Chips Act)

Target Audiences

European Commission & Funding Bodies: Use the 4-Filter Prioritization Framework to allocate Horizon Europe, Digital Europe Programme, and ECCC funding to capabilities with the highest strategic impact.

National Cybersecurity Agencies: Leverage the progression matrices to coordinate national R&I investments with EU-wide timelines and ensure interoperability across Member States.

Research Consortia & Industry: Utilize the 7-Dimensional Analysis template to structure project proposals, clearly articulating baseline, target, drivers, risks, and alignment with European strategic goals.

Standardization Bodies (CEN/CENELEC, ETSI): Reference the architectural framework and MIMs to develop harmonized standards for secure, interoperable computing continuum infrastructure.

How to Navigate This Document

1. **Start with Section 1-3:** Understand the threat landscape, regulatory drivers, and strategic context for securing the European computing continuum.
2. **Review Section 4:** Familiarize yourself with the 6 Security Layers and 4 Cross-Layers comprising the architectural framework.
3. **Apply Section 5:** Use the 4-Filter Prioritization Framework to assess which capabilities are critical for your specific context (national strategy, industrial sector, research focus).

4. **Study Section 6-7:** Understand the 7-Dimensional Analysis methodology and ODP/SSL progression metrics before designing projects or policies.
5. **Reference Section 9 (Priorities 1-3):** Use the detailed capability profiles as templates for structuring additional priorities relevant to your domain.
6. **Implement Section 10:** Actionable strategic recommendations for Phase 1 (2025-2030) provide concrete next steps.

Key Principles

- **Evidence-Based Prioritization:** All priorities must pass the 4-Filter Framework (Regulatory Urgency, Foundational Dependency, Sovereignty Cliff, Market Feasibility).
- **Sovereignty-First Design:** Investments should systematically reduce foreign dependencies (measured via SSL metrics) while building European industrial capacity.
- **Compliance-by-Design:** All R&I outputs must align with or anticipate NIS2, CRA, AI Act, and Data Act requirements to ensure market relevance.
- **Federated Interoperability:** Solutions must prioritize standardization and open architectures (e.g., Simpl middleware, open-source RISC-V) over proprietary, siloed approaches.
- **Measurable Progression:** Avoid subjective TRL assessments; instead, use objective ODP (market integration) and SSL (sovereignty) metrics with clear phase transition criteria.

2 Background & Driving Factors

Destinations:

- **Primary:** 4. A secure, sovereign European computing continuum infrastructure.
- **Secondary:** 6. European leadership in emerging disruptive computing paradigms.
- **Decentralization Risk:** The paradigm shift from the centralized Cloud to a distributed Cloud-Edge-IoT continuum radically expands the attack surface. This creates environments that are often “too many, too diverse, and too dispersed” to secure with traditional perimeter defenses.
- **Physical Exposure:** Edge nodes and far-edge IoT devices frequently operate in the field. Unlike hyperscaler data centers, they lack strict physical and environmental security controls, making them highly vulnerable to tampering.
- **Regulatory Pressure:** There is an immediate, complex need to operationalize the NIS2 Directive [7], the Cyber Resilience Act (CRA) [8], and the AI Act [9] across a highly fragmented technological ecosystem to ensure compliance and market access.
- **Digital Sovereignty:** Europe must urgently reduce its systemic reliance on non-EU technology providers. Doing so will mitigate extra-territorial data access risks and avoid geopolitical vendor lock-in, ensuring independent control over critical infrastructure.

2.1 Europe in Relation to the State of the Art

Comparison vs. Hyperscalers:

- **EU Strengths:** Europe leads globally in Trust, Privacy-by-Design, proactive regulation (GDPR, Data Act, AI Act), and rigorous, standardized certification frameworks (e.g., EUCS [10]).
- **EU Gaps:** The European market suffers from a lack of unified, hyper-scale security orchestration. Furthermore, there is a critical dependency on non-EU hardware (secure silicon) and highly fragmented threat intelligence sharing among member states.

Strategic Goal: To definitively transition the European computing continuum from a state of “reactive, siloed security” to an advanced architecture defined by “autonomous, federated, and certified Security-by-Design.”

3 Related Work

Existing research on cybersecurity for distributed computing infrastructures has primarily focused on isolated domains such as cloud security, edge computing protection, and IoT security frameworks. However, limited work addresses the holistic security challenges of the integrated Cloud–Edge–IoT continuum.

Recent initiatives, including the European Union Cybersecurity Strategy, ENISA reports, and Horizon Europe projects, emphasize the need for sovereignty-driven security architectures. Prior studies on Zero Trust Architectures (ZTA), Post-Quantum Cryptography (PQC), and confidential computing provide important building blocks but lack a unified prioritization and transition methodology.

This work differentiates itself by proposing a structured, end-to-end methodological pipeline that connects architectural capabilities with policy-driven prioritization and phased implementation strategies.

4 Related Work and Methodological Positioning

This section surveys the three research traditions whose synthesis motivates the methodology presented in Section 5: technology roadmapping (TRM), capability-based planning (CBP), and cybersecurity capability maturity assessment. It then reviews the methodological approaches of all major European cybersecurity roadmap deliverables and establishes the specific gap that the present work addresses.

4.1 Technology Roadmapping Foundations

Technology roadmapping provides the procedural backbone for structured strategic planning under technological uncertainty. The Cambridge school, anchored by Phaal, Farrukh, and Probert [1], formalises roadmapping as a series of facilitated workshops progressing through *Market* → *Product* → *Technology* → *Charting*, with each workshop gated by the outputs of its predecessor. The accompanying T-Plan workbook [11] operationalises this into an industry-applicable fast-start process. The parallel Sandia tradition, codified by Garcia and Bray [2], structures roadmapping as a three-phase process (*Preliminary activity* → *Development* → *Follow-up*) with seven development steps progressing from product focus through critical requirements, technology areas, drivers, alternatives, and timelines to recommendations. Kostoff and Schaller [12] provide the first formal taxonomy of roadmapping approaches in the engineering management literature, distinguishing expert-based from bibliometric methods and defining the node-link structure inherited by subsequent work.

The canonical TRM vocabulary organises content into three layers: *drivers* (why), *products/services/capabilities* (what), and *technologies/resources* (how), plotted against time on a generic chart [1, 13]. Critically, “capability” is an optional middle-layer label in Phaal’s formulation, not an organising primitive with a formal definition. Terms such as “focus areas” and “priorities” are derivative: priorities are rankings applied to drivers or gaps, and focus areas are thematic clusters used loosely in government and cybersecurity contexts but absent from the Cambridge and Sandia lexicon.

Two observations are significant for the present work. First, no canonical TRM framework separates “capability identification” from “capability analysis” as distinct named stages. Garcia and

Bray [2] fold both into their Development phase; T-Plan separates identification (workshops 1–3) from integration (workshop 4) but labels both as “development.” The explicit separation of these functions into named pipeline stages is a methodological design choice justified in Section 5. Second, quantitative maturity within TRM relies predominantly on Technology Readiness Levels (TRL) [14,15], with extensions including Manufacturing Readiness Levels (MRL) and System Readiness Levels (SRL) [16]. Cambridge TRM is predominantly qualitative; TRL-annotated roadmaps are more common in defence and aerospace applications. In cybersecurity specifically, quantitative maturity typically employs Maturity Indicator Levels (MIL 0–3) in the DOE Cybersecurity Capability Maturity Model (C2M2) [17] or Tiers 1–4 in the NIST Cybersecurity Framework [18], rather than TRL.

Stage-transition logic in TRM draws on Cooper’s Stage-Gate systems [19], which define gates as quality-control checkpoints comprising deliverables, must-meet and should-meet criteria, and Go/Kill/Hold/Recycle outputs. Gerdri, Vatananan, and Dansamasatid [20] provide the most operational transition logic in dynamic TRM, gating three stages (Initiation → Development → Integration) by explicit criteria: stakeholder acceptance, completion of information collection, and incorporation into ongoing planning. Carvalho, Fleury, and Lopes [21] identify recurrent quality axes for roadmap assessment—completeness, internal consistency, traceability, feasibility, and stakeholder legitimacy—that inform the gate criteria adopted in the present methodology.

4.2 Capability-Based Planning in Security Contexts

Capability-based planning (CBP) originates in defence doctrine. The canonical academic reference is Davis [3], who defines CBP as “planning, under uncertainty, to provide capabilities suitable for a wide range of modern-day challenges.” The operational implementations are the NATO Bi-SC Capability Codes and Capability Statements [22], which articulate capability requirements along the DOTMLPFI axes (Doctrine, Organisation, Training, Matériel, Leadership, Personnel, Facilities, Interoperability), and the US Department of Defense Joint Capabilities Integration and Development System (JCIDS) [23], which chains Capabilities-Based Assessment through Initial Capabilities Document, Capability Development Document, and Capability Production Document with DOTMLPF-P change recommendations.

The terminological distinctions within CBP are precise and methodologically significant. A *capability* is an outcome-oriented, solution-agnostic ability to achieve a desired effect. A *technology* is one of the means (the “M” in DOTMLPF). A *priority* is a ranking applied to capabilities or gaps. A *focus area* is a policy-level thematic cluster at a higher abstraction level than a capability. A *requirement* is a testable performance attribute that implements a capability. Davis’s central methodological warning—avoid “jumping to the solution”—is the logic that justifies separating capability identification from capability analysis in any structured planning pipeline.

CBP has crossed into cybersecurity only partially. Lehto and Limnell [24] apply DOTMLPF-I in cyber defence contexts. AlDaajeh and Alrabae [25] synthesise seven national cybersecurity strategy axes (legal, technical, organisational, capacity-building, cooperation, economic, socio-cultural) and AlDaajeh *et al.* [26] map national cyber strategies onto capability objectives. Curtis and Mehravari [27] bridge CERT resilience management with capability maturity. However, no widely cited publication imports the full Davis/JCIDS analytical machinery into civilian cybersecurity roadmapping. In established cyber frameworks, “capability” is used loosely: the NIST CSF 2.0 [18] employs *Functions*, *Categories*, *Subcategories*, *Outcomes*; the NICE framework [28] uses *Tasks*, *Knowledge*, *Skills*, *Work Roles*; the MITRE ATT&CK framework is an adversary-behaviour taxon-

omy (*Tactics, Techniques, Procedures*), not a defender-capability taxonomy; and ENISA’s National Capabilities Assessment Framework [29] uses “capability” closer to a maturity domain than to Davis’s outcome-oriented definition. The present work adopts Davis’s definition explicitly, grounding the terminology in the CBP tradition to achieve greater precision than prior EU cyber roadmaps.

4.3 Multi-Dimensional Capability Assessment Frameworks

No single peer-reviewed framework combines technical readiness, regulatory alignment, market dynamics, and sovereignty assessment in a unified per-capability template. The relevant contributions are distributed across four disciplines.

The defence tradition supplies DOTMLPF-P and DOTMLPFI, providing eight-axis internal decomposition of capability solutions but lacking external market and regulatory axes. The strategic management tradition supplies PESTEL (Political, Economic, Social, Technological, Environmental, Legal), recently applied to cybersecurity by Morgado *et al.* [30]. The engineering tradition supplies the readiness-level suite: TRL [14], MRL, IRL, and SRL [16], reviewed systematically by Olechowski, Eppinger, and Joglekar [31]. These readiness scales are strictly technical; they do not natively encode regulatory, market, or sovereignty dimensions. The political science tradition supplies the digital sovereignty axis, with canonical references including Pohle and Thiel [32], Roberts *et al.* [33] (defining five sovereignty sub-dimensions: data governance, platform power, digital infrastructure, emerging technology, and cybersecurity), Celeste [34], and Falkner *et al.* [35]. None of these constitutes a per-capability profiling template.

Cybersecurity maturity models come closest to multi-dimensional profiling. The Oxford Global Cyber Security Capacity Centre Maturity Model for Nations (GCSCC CMM) [36] uses five dimensions, 24 factors, and five maturity stages (start-up through dynamic). Structurally, it follows a four-step logical pattern—framework identification, capability enumeration, profiling, phased progression—that closely parallels the pipeline proposed in this paper, but it is applied to national-level capacity assessment rather than technical cybersecurity capabilities and does not anchor to architectural reference models. The DOE C2M2 [17] employs ten domains with approximately 356 practices scored at MIL 0–3. The NIST CSF 2.0 [18] uses six Functions, Categories, Subcategories, and four implementation Tiers. The ITU Global Cybersecurity Index uses five pillars (legal, technical, organisational, capacity development, cooperation). AlDaajeh and Alrabaaee [25] synthesise seven national strategy axes but lack sovereignty and readiness dimensions.

The gap is clear: GCSCC covers technical, regulatory, societal, educational, and strategic aspects, but sovereignty and market dynamics are only implicit. ENISA’s National Capabilities Assessment Framework [29] covers regulatory and sectoral maturity but omits TRL and market axes. Roberts *et al.* [33] add sovereignty but at the policy-discourse level, not as a per-capability metric. A profiling template that fuses technical readiness, regulatory alignment, market feasibility, and an explicit digital sovereignty axis within a single per-capability instrument—as proposed in the 7-Dimensional Capability Profile of Section ??—has no direct precedent in the surveyed literature.

4.4 European Cybersecurity Roadmap Precedents

Every major European cybersecurity roadmap deliverable produced under the Horizon 2020 Cybersecurity Competence Network pilots has been reviewed. A consistent observation emerges: none cites the canonical TRM literature (Phaal, Garcia & Bray, Kostoff) directly, and none employs quantitative per-capability maturity metrics.

The sole EU precedent that formally grounds cybersecurity roadmapping in TRM theory is the CyberROAD FP7 project. Ariu *et al.* [4] cite Phaal, Probert, Carvalho, and the EIRMA formulation, proposing a co-evolutionary model of technology, threat, and policy trajectories. CyberROAD omits architecture-anchored capability derivation, regulatory and sovereignty profiling axes, and quantitative maturity progression. It is the single most important methodological predecessor to the present work.

The four H2020 Competence Network pilots each contribute distinct methodological elements. CONCORDIA [5] (Deliverable D4.4, 2022) organises its roadmap across six dimensions (Research & Innovation; Education & Skills; Legal & Policy; Economics & Investments; Certification & Standardisation; Community Building) with a per-dimension narrative pipeline from threat landscape through challenge identification to phased recommendations across three time horizons. It offers the strongest sovereignty-aware multi-dimensional framing among the pilots but employs no quantitative maturity metrics. SPARTA [37] (Deliverables D3.1–D3.5, 2019–2022) structures challenges against the JRC Cybersecurity Taxonomy (domains $\times$ sectors) with a per-challenge template (Final Goal, EU Benefits, JRC Domain, JRC Sector, Emerging Technology relation). The taxonomy-grounded derivation is a strength, but quantitative maturity tracking is absent. CyberSec4Europe [6] (Deliverable D4.5, 2022) is the strongest multi-dimensional precedent: a structured per-vertical schema across seven application verticals profiles each challenge across SWOT analysis combined with Digital Sovereignty, Green Deal, and Digital Decade Strategy alignment, phased into three time horizons. Its methodology section explicitly paraphrases Phaal’s formulation of roadmaps as “structured visual representations.” ECHO [38] (Deliverables D4.3 and D4.10) takes a methodological outlier path by mapping SCRUM onto roadmap development—EPICs, User Stories, acceptance criteria—and adds Analytic Hierarchy Process (AHP)-based multi-sector assessment. Quantitative prioritisation is present, but no unified phased-maturity matrix is produced.

Beyond the Competence Network, the European Cyber Security Organisation (ECSO) Strategic Research and Innovation Agenda [39] uses SWOT analysis across thematic pillars without per-priority multi-dimensional templates or maturity metrics. ENISA’s Threat Landscape Methodology [40] supplies the most explicit multi-stage analytical pipeline in EU cybersecurity—six steps (Direction $\rightarrow$ Collection $\rightarrow$ Processing $\rightarrow$ Analysis & Production $\rightarrow$ Dissemination $\rightarrow$ Feedback)—but adapts the classical intelligence cycle, not TRM, and is threat-centric rather than capability-centric. The ECCC Strategic Agenda [41] is a policy strategy rather than an operational roadmap methodology.

Table 1 summarises the terminological heterogeneity across EU cybersecurity roadmaps, illustrating that “capability” is almost never employed as the formal primary unit of analysis.

4.5 Gap Analysis and Methodological Positioning

The four research streams surveyed above are consolidated against the four pipeline elements that define the present methodology: (a) architecture-anchored capability derivation, (b) capability enumeration from architecture, (c) multi-dimensional profiling covering technical, regulatory, market, and sovereignty axes, and (d) phased progression matrix with quantitative maturity metrics. Table 2 presents the assessment.

No existing framework integrates all four pipeline elements. CyberSec4Europe D4.5 is the closest precedent on multi-dimensional profiling; ECHO is the closest on quantitative prioritisation via AHP; ENISA’s Threat Landscape methodology is the closest on explicit multi-stage analytical pipelines; CONCORDIA is the closest on sovereignty-driven framing; and the GCSCC CMM is

Table 1: Primary analytical units in EU cybersecurity roadmaps

Project	Primary Unit	Grouping
CONCORDIA	Recommendations, challenges	Dimensions
ECSO	Priority areas, challenges	Pillars, WGs
ENISA	Threats, incidents	Landscapes
SPARTA	Challenges	JRC domains
CyberSec4Europe	Research chal- lenges	Verticals
ECHO	User Stories	EPICs
ECCC	Investment priori- ties	Strategic pillars

Table 2: Gap analysis: existing frameworks assessed against the four pipeline elements of the proposed methodology. Entries indicate coverage: Yes, Partial, Weak, or No.

Framework	(a) Arch.	(b) Enum.	(c) Multi-dim.	(d) Quant.	All 4?
CyberROAD [4]	Partial	Yes	Partial	No	No
NIST CSF 2.0 [18]	Yes	Partial	Weak	Partial	No
ENISA NCAF [29]	Partial	Yes	Partial	Yes	No
DOE C2M2 [17]	No	Yes	No	Yes	No
Phaal T-Plan [1]	Generic	Yes	Partial	Possible	No
GCSCC CMM [36]	No	Yes	Partial	Yes	No
CONCORDIA D4.4 [5]	Yes	Yes	Partial	No	No
SPARTA D3.4 [37]	Yes	Yes	Partial	No	No
CyberSec4Europe D4.5 [6]	Yes	Yes	Strong	No	No
ECHO D4.3 [38]	Yes	Yes	Partial	Partial	No
This work	Yes	Yes	Yes	Yes	Yes

the closest structural analogue but operates as a national-capacity maturity model rather than a technical roadmap anchored to architectural reference models.

The present methodology occupies this gap by fusing three traditions: the Cambridge/Sandia TRM tradition [1,2] for phased planning structure and stage-gate logic, the Davis/JCIDS capability-based planning tradition [3] for formal capability definition and architecture-to-capability derivation, and cybersecurity maturity models [17,18,36] for quantitative progression metrics. The novelty is defensible on three independent grounds: (i) no canonical TRM framework separates capability identification from capability analysis as named, gated stages; (ii) no EU cybersecurity roadmap employs quantitative per-capability maturity metrics anchored to an architectural reference model; and (iii) no peer-reviewed framework profiles capabilities simultaneously across technical readiness, regulatory alignment, market dynamics, and digital sovereignty in a single template.

5 Methodology

The methodology presented in this section is designed to transform a broad and heterogeneous strategic landscape—spanning hardware security, post-quantum cryptography, AI-driven defence, federated governance, and supply chain integrity across the European cloud–edge–IoT computing continuum—into a structured roadmap for research, innovation, policy, deployment, and ecosystem development. The analytical process proceeds through three staged phases: architectural decomposition, capability formulation and profiling, and priority clustering with phased progression, followed by an iterative validation step.

Two foundational commitments shape the methodology. First, it is intended to support strategic roadmap construction, not deterministic forecasting. In complex socio-technical ecosystems, progress is non-linear and depends on interacting technical, regulatory, operational, and market developments whose timing and interdependencies cannot be predicted with precision. Second, the methodology is designed to connect technical cybersecurity capabilities with broader European strategic concerns—resilience, interoperability, trust, and what we term *Sovereignty Level*: the degree to which critical capabilities, control points, and dependencies remain under credible European influence, governance, or operational control. Sovereignty Level is employed throughout the methodology as an interpretive strategic lens rather than as a rigid scoring model; it guides assessment of whether progress in cybersecurity and cyber-resilience strengthens Europe’s strategic control, trusted autonomy, and reduced dependency in critical parts of the computing continuum.

5.1 Stage 1: Core Dimensions and Focus Area Identification

The first stage defines the analytical scope and structure of the roadmap by decomposing the cloud–edge–IoT computing continuum into a set of architectural layers and cross-cutting concerns. These dimensions are treated as analytically distinct but operationally interdependent: a vulnerability or governance gap at one layer typically has consequences that propagate to adjacent layers, and effective security requires coordinated action across the full vertical and horizontal extent of the architecture.

The purpose of this stage is to answer three questions. Where are the major security and resilience problem domains within the distributed computing continuum? At what architectural layers or cross-layers do these problems emerge? And which focus areas are sufficiently strategic, cross-cutting, or underdeveloped to justify dedicated roadmap attention?

Focus areas are derived from four complementary sources. The first is the architectural characteristics of distributed cloud–edge–IoT systems themselves: the heterogeneity of execution environments spanning HPC, cloud, edge, and device tiers creates structurally distinct security requirements at each level of the stack [42]. The second source is the set of recurring cybersecurity and cyber-resilience challenges observed across these tiers—hardware integrity [?], software supply chain verification, workload isolation, identity federation, data confidentiality, and network perimeter control [43]. The third comprises cross-cutting strategic concerns that cannot be localised to a single layer: sovereignty, trust, interoperability, resilience, assurance, and regulatory alignment—driven by the simultaneous enforcement of NIS2 [7], the Cyber Resilience Act [8], the AI Act [9], and the Digital Operational Resilience Act [44]. The fourth is the practical need to connect technical architecture with operational and governance realities—recognising that a capability with no credible deployment pathway or regulatory anchor is unlikely to achieve ecosystem-wide impact regardless of its technical merit.

The output of Stage 1 is not a priority list but a structured map of the problem space. This map is organised into six Security Layers (the vertical stack) and four Transversal Cross-Layers (the horizontal enablers), as illustrated in the architectural framework (Section ??). The six Security Layers address: (i) hardware-centric security, (ii) software-centric security at the middleware and operating system level, (iii) application-centric security for workloads and services, (iv) user and orchestration-centric security including identity and access management, (v) data-centric security encompassing confidential computing and cryptographic protection, and (vi) network infrastructure and perimeter security including telco-cloud, non-terrestrial networks, and physical resilience. The four Cross-Layers address: (CL1) automated security processes and the cyber-resilient AI lifecycle, (CL2) dynamic risk management and operational cyber resilience, (CL3) interoperable security mechanisms and open-source verifiability, and (CL4) compliance-by-design with agile governance and automated assurance.

Within this structured map, some focus areas acquire greater strategic importance because they influence Europe’s future Sovereignty Level. This is particularly the case where dependencies on external providers, infrastructures, fabrication capacity, toolchains, or assurance mechanisms are structurally significant—for example, the near-complete European dependency on non-EU foundries for secure silicon fabrication, or the dominance of non-EU vendors in the operational trusted execution environment market [42]. Stage 1 identifies these sovereignty-sensitive focus areas explicitly so that subsequent stages can track their evolution as a strategic concern alongside purely technical maturity.

5.2 Stage 2: Capability Identification and Profiling

Once the architectural map and its constituent focus areas are defined, the methodology moves from problem framing to capability formulation. A *capability*, in the context of this roadmap, is understood as a strategic ability that the European ecosystem should be able to design, deploy, govern, standardise, assure, coordinate, or scale in order to achieve trustworthy, resilient, and sovereign operation of the computing continuum. Capabilities are not limited to technical controls; they may also encompass governance mechanisms, interoperability instruments, assurance functions, ecosystem coordination capacities, and deployment enablers.

Capabilities are derived by translating each focus area identified in Stage 1 into actionable strategic needs. This translation is guided by a consistent question: what must Europe be able to do—technically, institutionally, and industrially—to address this focus area at the scale and with the assurance level that the regulatory and threat environment demands? The ten capabilities identified through this process (detailed in Section ??) span the full range from hardware roots of trust and post-quantum cryptographic agility to federated security interoperability and proactive cyber threat management.

5.2.1 The Seven-Dimension Capability Profile

Each capability is assessed through a seven-dimension analytical template applied identically across all priorities. This template ensures that every capability is characterised not only from a technical perspective, but also in terms of its regulatory alignment, investment pathway, failure modes, and contribution to European strategic autonomy. The consistency of the template enables structured cross-priority comparison—a feature essential both for funding prioritisation and for identifying shared dependencies and bottlenecks that span multiple capabilities.

Dimension 1: Baseline and Target Objective. This dimension grounds the capability in reality by describing the present condition of the European ecosystem with respect to the capability domain, the intended future condition, and the strategic change the capability is meant to achieve. The baseline captures current industrial capacity, the degree of reliance on non-European vendors, technical and economic barriers to adoption, and the state of standardisation and certification. The target defines measurable end-states—quantified market share objectives, certification levels, migration coverage metrics, or deployment thresholds—that serve as the basis for evaluating progress and structuring funding calls in Stage 3.

Dimension 2: Primary Drivers. This dimension articulates the external pressures that make the capability urgent and non-deferrable. It distinguishes between two categories. *Threat drivers* identify adversarial capabilities or attack vectors that are already active or imminently scalable—for example, state-sponsored exploitation of hardware debug interfaces on field-deployed edge nodes, the “Harvest Now, Decrypt Later” strategy enabled by advances in quantum computing [45], or the weaponisation of generative AI for polymorphic malware production. *Regulatory drivers* identify specific EU legal obligations with enforcement timelines, citing the precise directive or regulation and its relevant provisions—for example, NIS2 Article 21(2)(e) mandating cryptographic policies for essential entities [7], or CRA Articles 10–11 imposing hardware security certification requirements with enforcement anticipated in 2027 [8].

Dimension 3: High-Impact Activities. This dimension identifies three to five concrete, actionable research and innovation activities within the capability domain that are expected to yield the greatest strategic return on investment. Each activity is formulated as a specific, measurable initiative rather than a general aspiration, and is tagged with its primary target phase (Phase 1, Phase 2, or Phase 3). This phasing ensures that the transition matrix constructed in Stage 3 is anchored in concrete, fundable work rather than in abstract descriptions of desired outcomes.

Dimension 4: Catalysts and Enablers. This dimension identifies the concrete mechanisms, funding vehicles, and regulatory standardisations required from the European institutional ecosystem to accelerate the capability from its baseline toward its target state. Three categories of enablers are distinguished. *Research and funding alignment* identifies relevant Horizon Europe clusters, Digital Europe Programme calls, and European Cybersecurity Competence Centre (ECCC) Strategic Agenda work packages [41]. *Regulatory and standardisation* identifies required CEN/CENELEC, ETSI, or ISO standards, distinguishing clearly between existing standards that should be referenced and new standards that must be developed. *Ecosystem enablers* identify pilot infrastructures such as federated cyber ranges [5], industrial partnerships such as those coordinated through Gaia-X or IPCEI-CIS, and capacity-building programmes—particularly those enabling SME participation without prohibitive certification or compliance costs.

Dimension 5: Disruptors and Systemic Risks. This dimension acknowledges that research and innovation trajectories are non-linear by identifying risks that could render planned deployments obsolete or unachievable. Three risk categories are assessed. *Technical cliffs* are points at which a defensive capability is rendered ineffective by a technological or adversarial breakthrough before EU countermeasures are fully deployed—for example, the arrival of cryptographically relevant quantum computers before post-quantum migration is complete [45], or the discovery of a new side-channel attack class that bypasses current trusted execution environments [46]. *Supply*

chain and market risks encompass fabrication bottlenecks, vendor resistance to open standards, monopolisation of critical components by non-EU providers, or SME exclusion from procurement pathways. *Geopolitical and regulatory risks* cover export controls on critical components, extra-territorial jurisdiction conflicts, or fragmented national implementations that undermine EU-wide interoperability.

Dimension 6: Failsafe Mechanisms and Resilience. This dimension defines cyber-resilience as applied to the roadmap itself by specifying engineering fallbacks if a capability’s primary mechanism is compromised, delayed, or only partially realised. Four categories of failsafe are addressed. *Hybrid transitional schemes* involve parallel deployment of the target technology alongside incumbent solutions during migration windows—for example, classical-plus-post-quantum hybrid cryptographic schemes as specified in ETSI TS 103 744 [47]. *Graceful degradation* defines the minimum viable operational mode the infrastructure can sustain if the primary capability becomes unavailable—for example, cached-credential or local-only operation during connectivity loss. *Autonomous isolation protocols* specify automated quarantine, revocation, or containment mechanisms that activate when a component fails attestation, preventing compromise propagation without manual intervention. *Implementation diversity* mandates deliberate multi-vendor or multi-algorithm diversity to avoid monoculture risks—for example, deploying multiple TEE architectures (Intel TDX, AMD SEV-SNP, ARM CCA, RISC-V Keystone [46]) or multiple post-quantum algorithm families (ML-KEM [48], ML-DSA [49], SLH-DSA [50]) rather than concentrating on a single vendor or standard.

Dimension 7: Transversal Meta-Layer (Policy Alignment). This dimension elevates the technical capability to the political and societal level, linking it to the EU’s overarching strategic autonomy agenda. It assesses the capability’s contribution across three policy axes. *Sovereignty and autonomy*: the specific foreign dependencies the capability reduces and, where data permits, the current dependency level. *Trust and ethics*: alignment with GDPR principles [51], Privacy-by-Design, algorithmic transparency, and trustworthiness requirements under the AI Act [9]. *Sustainable security*: the energy efficiency of the proposed cryptographic or monitoring operations, reduction of electronic waste through crypto-agility and software-defined upgrades, and alignment with EU Green Deal objectives.

5.2.2 Stage 2 Output

For each capability, Stage 2 produces a complete seven-dimension profile. The baseline (Dimension 1) and high-impact activities (Dimension 3) directly inform the operational characterisation of each phase in the Stage 3 transition matrix. Catalysts and enablers (Dimension 4) populate the transition conditions that mark phase boundaries. Disruptors (Dimension 5) and failsafe mechanisms (Dimension 6) inform the resilience provisions in the cross-cutting enablers analysis and the executive summary. The seven-dimension template thus serves as the analytical bridge between the architectural decomposition of Stage 1 and the phased strategic synthesis of Stage 3.

5.3 Stage 3: Priority Clustering and Transition Matrix

Not every capability can or should be treated as a standalone roadmap priority. Capabilities that share architectural proximity, common enabling conditions, similar policy or standardisation needs,

overlapping ecosystem actors, related deployment trajectories, or convergent strategic outcomes benefit from coordinated treatment. Stage 3 therefore groups the ten capabilities identified in Stage 2 into three broader Priority Clusters that reflect these shared properties and enable a more tractable strategic analysis.

The roadmap is organised into the following three Priority Clusters:

Priority Cluster 1: Sovereign Trusted Infrastructure — the hardware, cryptographic, and resilience foundations that anchor trust across the continuum.

Priority Cluster 2: Cognitive Security and Autonomous Cyber Defence — the AI-driven operational layer that scales defence beyond human-paced response.

Priority Cluster 3: Federated Operational Sovereignty — the interoperability, supply-chain, and federation substrate that enables a unified European trust fabric.

Each cluster is analysed through a three-phase transition matrix:

Phase 1: Near-Term Implementation and Pilot Actions (2026–2030) — deploying near-ready solutions, enforcing enacted regulation, and closing governance gaps before technical work scales.

Phase 2: Mid-Term Scale-Up, Industrial R&D and Standardisation (2030–2035) — industrialising sovereign capabilities, federating architectures, and building the workforce and institutional infrastructure the technical work requires.

Phase 3: Long-Term Strategic, Frontier R&I and Leadership (2035–2040) — achieving full operational maturity, pioneering frontier research, and resolving the structural doctrine and oversight questions that earlier phases cannot settle alone.

5.3.1 Analytical Perspectives

For each priority cluster and phase, the assessment considers four analytical perspectives.

R&I maturity captures where the constituent capabilities stand in terms of research, development, demonstration, and readiness for industrial deployment. This perspective tracks whether the technical foundations are established, whether proof-of-concept has been achieved, and whether industrialisation is feasible within the phase timeframe.

Policy, regulation, and standards assesses the state of the regulatory and standardisation environment—whether enabling legislation has been enacted, whether enforcement timelines are credible, whether the necessary standards exist or must still be developed, and whether conformity assessment infrastructure is operational.

Operational deployment and adoption evaluates the extent to which capabilities have moved beyond laboratory or pilot environments into operational use by infrastructure operators, service providers, and end-user organisations, including the specific actors involved and the approximate scale of deployment.

Market uptake and strategic impact considers whether the capability is generating commercial traction, whether European providers are competitive, and whether the capability is contributing to a measurable improvement in Sovereignty Level—for instance, through reduced critical external dependencies, stronger European operational control, improved assurance capacity, or strategic industrial positioning.

The matrix is intended to show how each cluster evolves over time across these four perspectives, rather than to provide a static list of recommendations. This dynamic representation acknowledges that the most important strategic information often lies in the interdependencies between clusters and between phases—for instance, the fact that Cluster 2’s autonomous defence capabilities gain operational credibility only in proportion to the maturity of Cluster 1’s hardware and cryptographic substrate [42].

5.3.2 Transition Conditions

A critical methodological choice concerns the treatment of boundaries between phases. This roadmap uses *transition conditions* rather than hard gates. Transition conditions are evidence-informed progression markers indicating when a priority cluster appears sufficiently prepared to move from one phase to the next. They are not binary thresholds or deterministic requirements. They are framed as analytical signals of increasing readiness, coherence, and ecosystem support.

This design choice reflects four methodological purposes. First, transition conditions make the roadmap temporally structured without imposing false precision. Second, they connect research progress with deployment and governance realities, ensuring that the roadmap does not treat technical maturity in isolation from the institutional and market conditions required for real-world impact. Third, they avoid the rigidity of deterministic forecasting in a domain where adversarial breakthroughs, regulatory delays, fabrication bottlenecks, and geopolitical disruptions can alter timelines unpredictably. Fourth, they support iterative updating: as evidence accumulates and the ecosystem evolves, transition conditions can be reassessed without invalidating the overall roadmap structure.

Definition. A transition condition is a short statement indicating that progress has been reached in one or more of the following areas: a technical basis has become sufficiently mature for the next stage of deployment; a policy or standardisation basis has emerged that reduces adoption uncertainty; a deployment pathway has been validated through pilots or integration efforts; a strategic ecosystem condition has materialised, such as industrial commitment, interoperability support, assurance capacity, or supply-side readiness; or a Sovereignty Level improvement has become plausible through reduced dependency, stronger control, or increased European capacity in a strategically relevant domain.

Form. Each transition condition is written as a single sentence following a structure such as: “*Advancement to the next phase is indicated when there is credible evidence of [readiness marker]*”—or equivalently—“*[Cluster or capability progression] becomes plausible when [technical / policy / deployment / ecosystem / sovereignty-related condition] is sufficiently established.*”

Illustrative examples. Well-formed transition conditions include statements such as: a common baseline for interoperability, assurance, or certification has been defined and is referenced by relevant actors; pilot deployments have demonstrated feasibility beyond laboratory conditions in representative operational environments; a regulatory or standards pathway has become sufficiently clear to reduce adoption uncertainty for industrial actors; tooling, integration methods, or assurance practices have matured enough to support repeatable, non-bespoke deployment; dependencies on non-European critical components are partially mitigated through credible alternatives, diversi-

fication, or strengthened European control; or European actors have acquired sufficient operational or governance capacity to influence the strategic direction of the capability domain.

Scope limitations. Transition conditions are not promises, deterministic predictions, or precise quantitative thresholds unless evidence genuinely supports them. They do not claim that all actors move at the same speed or that progression is irreversible. They should be interpreted as structured indicators of directional readiness—useful for strategic planning and coordination, but subject to reassessment as the ecosystem evolves.

5.4 Validation: Expert Review, Consultation, and Iterative Refinement

The roadmap is not treated as a closed analytical product but as an evolving strategic artefact that benefits from expert review, stakeholder consultation, and iterative revision. The methodology therefore includes a validation step intended to improve the robustness, relevance, and legitimacy of the roadmap’s analytical outputs.

Validation addresses five areas. First, whether the architectural layers and focus areas identified in Stage 1 are complete and relevant—that is, whether the problem space decomposition adequately covers the security and resilience challenges of the European cloud–edge–IoT continuum without material omissions or redundancies. Second, whether the capabilities formulated in Stage 2 are well-defined, actionable, and non-overlapping—ensuring that each capability represents a strategically distinct need rather than a restatement of another capability at a different level of abstraction. Third, whether the priority clustering in Stage 3 is coherent—that is, whether the grouping of capabilities into clusters reflects genuine shared dependencies and convergent implementation pathways rather than arbitrary aggregation. Fourth, whether the proposed transition conditions are credible, intelligible, and useful—specific enough to guide interpretation but broad enough to remain robust under the uncertainty inherent in a fifteen-year strategic horizon. Fifth, whether the implied Sovereignty Level assessments are reasonable and strategically meaningful—that is, whether the characterisation of European dependency, control, and capacity at each phase is grounded in observable evidence rather than aspiration.

This validation process draws on expert workshops conducted within the NexusForum and EuCloudEdgeIoT Working Group on Cybersecurity, stakeholder consultation with national cybersecurity agencies and industry representatives, working group review cycles, and policy dialogue with European Commission services. The roadmap is designed to accommodate future updates as the regulatory landscape evolves, as technical capabilities mature, and as the geopolitical and market context shifts.

5.5 Limitations and Methodological Positioning

The approach described above is intended to provide a transparent and defensible basis for roadmap construction in a complex and evolving ecosystem. It does not eliminate uncertainty. Instead, it organises uncertainty by using structured analytical stages, comparative capability profiling through the seven-dimension template, clustered priorities reflecting shared dependencies and implementation pathways, and transition conditions as indicative progression markers.

Several limitations should be acknowledged. The architectural decomposition, while comprehensive, reflects a particular analytical perspective and may not capture all relevant dimensions of the problem space—particularly those that emerge from the interaction of cybersecurity with adjacent

domains such as energy policy, industrial policy, or international trade. The seven-dimension profiles, while systematic, rely on expert judgement for assessments of baseline conditions, risk severity, and target feasibility; they do not constitute empirical measurement. The transition conditions, by design, sacrifice precision for robustness—they are better suited to strategic orientation than to operational planning at the individual project level. And the Sovereignty Level concept, while analytically useful, resists precise quantification: it is employed here as an interpretive strategic lens that helps assess whether progress strengthens Europe’s strategic control and reduced dependency, not as a metric with a defined scale.

These limitations are consistent with the roadmap’s intended use. The methodology is suitable for strategic agenda setting, expert discussion, cross-priority comparison, and iterative roadmap evolution—even when some dependencies, timings, or ecosystem shifts remain uncertain or contested. Its value lies not in predictive accuracy but in providing a shared analytical structure within which diverse stakeholders can coordinate research, policy, and investment decisions toward a common set of strategically defined objectives.

Security Framework with Separated Security Verticals

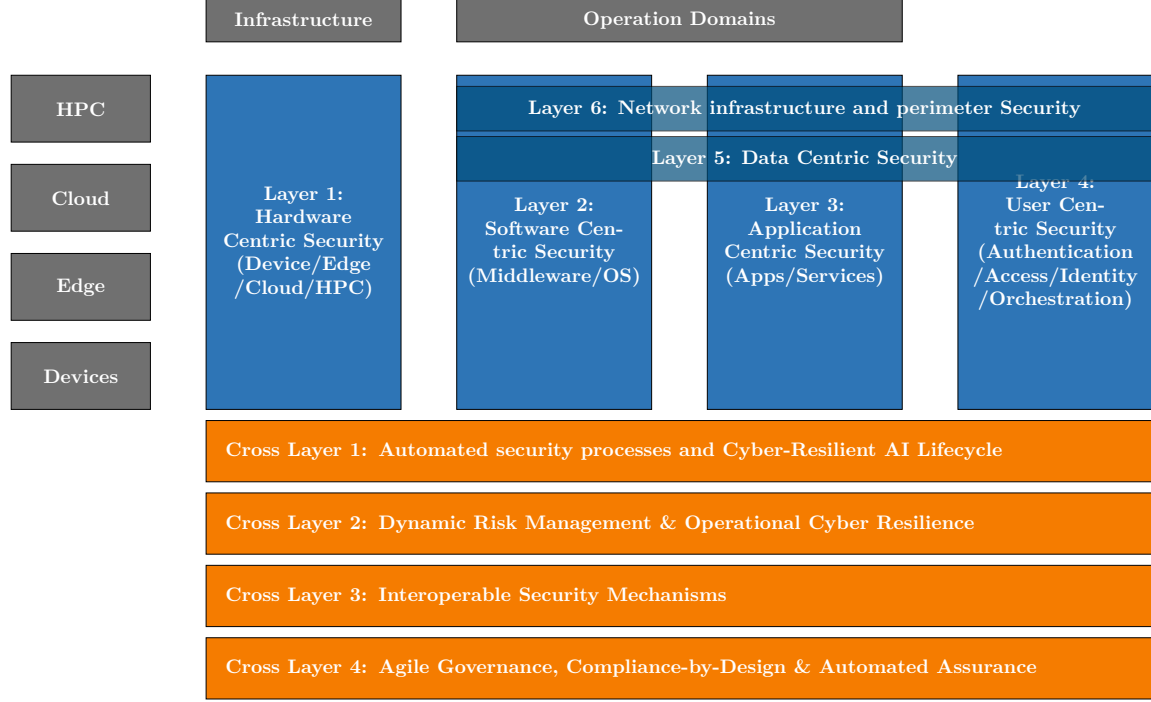


Figure 1: Security Framework with separated security verticals in dedicated boxes.

6 Capabilities (Architectural Framework)

As it is illustrated in Figure 1, the roadmap is structured across a cohesive architectural framework that spans four continuum environments (HPC, Cloud, Edge, Devices). The capabilities are organized into Six Security Layers (The Vertical Stack) and Four Transversal Cross-Layers (The Horizontal Enablers).

Part A: The Security Layers (Vertical Stack)

These layers define the specific technical security controls implemented across the computing infrastructure.

Layer 1: Hardware-Centric Security

1.1 Hardware Roots of Trust: Cryptographically verifiable supply chain trust (e.g., TPM attestation, secure enclaves) ensures physical device integrity. Success depends on driving European open-hardware designs (e.g., RISC-V) toward **EUCC 'High' assurance level certification** [52]

to mathematically guarantee resistance against state-level physical tampering and side-channel attacks at the vulnerable far-edge. **1.2 Physical-Layer (PHY) Threat Detection & Fingerprinting:** Exploiting intrinsic wireless channel characteristics, such as Channel State Information (CSI), Carrier Frequency Offset (CFO), and Received Signal Strength Indicator (RSSI). That is to establish real-time, keyless device authentication and anomaly detection at the lowest layer of the stack. This provides a hardware-anchored security primitive that is resilient even when upper-layer cryptographic processing is compromised or infeasible.

Layer 2: Software-Centric Security (Middleware/OS)

2.1 OS & Middleware Hardening: Securing hypervisors, container runtimes, and distributed operating systems is critical for isolation. This involves hardening KVM/QEMU and supporting sovereign, lightweight Linux distributions optimized for edge resource constraints, ensuring a reduced attack surface for the cognitive continuum.

2.2 Secure OTA Patch & Update Management: Edge nodes deployed in the field—smart grids, O-RAN radios, industrial gateways—require cryptographically verified, failure-resilient over-the-air (OTA) update pipelines. A compromised or failed update mechanism can render entire fleets of field-deployed nodes permanently vulnerable, making robust OTA management a critical software-layer capability.

Layer 3: Application-Centric Security (Apps/Services)

3.1 Workload Security: Securing containerized applications, microservices, and serverless functions executing dynamically across multi-provider nodes. This is achieved through hardened service meshes and sidecar proxy architectures that provide mutual TLS and granular traffic inspection without impacting performance.

3.2 In-Band Network Telemetry (INT) & Deep Observability: Embedding “smart telemetry” directly into the data packets traversing containerized workloads and service meshes, providing granular, real-time visibility into network behaviour and performance without requiring out-of-band monitoring infrastructure. This feeds high-fidelity signals to security analytics platforms.

3.3 API Security Gateway & Policy Enforcement: As containerized AI workloads increasingly communicate through REST/gRPC APIs, a dedicated API security gateway provides a unified enforcement point for authentication tokens, rate limiting, schema validation, and payload inspection—preventing injection attacks and data exfiltration at the application boundary.

Layer 4: User & Orchestration-Centric Security

4.1 Zero-Trust Orchestration: Seamless policy enforcement across a multi-provider computing continuum requires robust Policy Enforcement Point (PEP) and Policy Decision Point (PDP) architectures. This enables continuous authorization and dynamic access controls that adapt to the context of the request.

4.2 Federated Identity & Access Management (IAM): Cross-domain identity federation and verifiable credentials for machines, users, and automated workloads. Implementing **Self-Sovereign Identity (SSI)** principles allows for cryptographically secure, decentralized authentication across heterogeneous cloud-edge domains.

4.3 Privileged Access Management (PAM) & Just-In-Time (JIT) Access: Lateral movement by attackers frequently exploits over-provisioned standing privileges—the precise vector behind major Advanced Persistent Threat (APT) campaigns. PAM and JIT access controls ensure that elevated privileges are granted only for defined, time-bounded sessions with full auditability, minimizing the blast radius of any credential compromise.

Layer 5: Data-Centric Security

5.1 End-to-end Confidential Computing: Data-in-use security through Trusted Execution Environments (TEEs) spanning the entire Cloud-to-IoT data path. Leveraging TEE abstraction layers and remote attestation ensures that sensitive European data remains encrypted even during processing in non-trusted environments.

5.2 Post-Quantum & Cryptographic Agility: PQC standardization and hybrid crypto deployment are necessary to secure long-lived data. Priority is placed on **PQC migration for resource-constrained devices**, ensuring that edge infrastructure is resilient against “Harvest Now, Decrypt Later” quantum threats.

5.3 Signaling (Data) Security: Moving beyond best-effort packet forwarding by embedding trust directly into the network signaling architecture. This ensures the integrity and authenticity of control-plane messages, preventing hijacking of routing decisions and network configuration across the distributed continuum.

5.4 Distributed Reputation & Trust Networking: Implementing a ubiquitous, real-time reputation system built on Distributed Ledger Technology (DLT) and smart contracts. This enables autonomous, evidence-based trust decisions about nodes, services, and data sources across the continuum without reliance on a central authority.

5.5 Quantum Key Distribution (QKD) for Critical Links: While PQC addresses algorithmic quantum-resistance, QKD provides information-theoretically secure key exchange over optical fibre for point-to-point high-value links (government, critical infrastructure backbone). These two technologies are complementary and together form a comprehensive quantum-safe strategy.

5.6 Secure Multi-Party Computation (SMPC): Homomorphic Encryption and TEEs (Layer 5.1) handle confidential computing for single-party workloads. SMPC enables multiple mutually distrusting parties to jointly compute a function over their combined private inputs without revealing those inputs to each other—a critical capability for cross-border analytics and federated AI in regulated sectors.

Layer 6: Network Infrastructure & Perimeter-Centric Security

6.1 Telco-cloud & O-RAN Security: Hardening distributed 5G/6G cores and RAN components must align with the **Smart Networks and Services Joint Undertaking (SNS JU) SRIA** [53]. This ensures sovereign control over network slice isolation and virtualized RAN components.

6.2 Cross-Sector Physical Resilience & Safety: Adapting strict network controls and IT/OT gateway security for high-availability edge environments (e.g., Smart Grids). This guarantees that cyber-breaches in the digital layer do not cascade into physical safety hazards or kinetic damage in critical infrastructure.

6.3 Edge-Native Autonomous Guardians: Hosting lightweight, localized security control functions (Security-as-a-Service aggregators) directly at the far-edge gateways or within Open-RAN Radio Intelligent Controllers (RIC). These autonomous guardian nodes execute real-time

threat mitigation without requiring round-trip communication to a centralized cloud-based SOC, dramatically reducing response latency.

6.4 Non-Terrestrial Network (NTN) & Satellite Security: 6G will natively integrate satellite segments for coverage continuity, backhaul resilience, and emergency operations. These non-terrestrial links introduce distinct threat surfaces—including signal interception and orbital spoofing—requiring dedicated security requirements aligned with 3GPP NTN specifications and the EU Space Programme.

Part B: The Transversal Security Verticals (Cross-Layers)

These four cross-layers represent continuous, automated processes and governance structures that intersect every layer of the vertical stack.

Cross-Layer 1: Automated Security Processes & Cyber-Resilient AI Lifecycle

CL1.1 Continuous DevSecOps (Shift-Left): Embedding AI-driven code analysis and automated vulnerability remediation at the earliest stages of the software lifecycle.

CL1.2 Model & Pipeline Integrity (DataBOMs): Securing AI training pipelines to prevent data poisoning and unauthorized model manipulation.

CL1.3 Runtime AI Defense: Defending AI/ML infrastructure against real-world adversarial attacks (e.g., prompt injection, model evasion/theft) at the edge.

CL1.4 Dynamic Supply Chain Verification (Runtime SBOMs): Moving beyond static software manifests by implementing dynamic Software Bill of Materials (Runtime SBOMs) and “cryptographic proof of execution” to continuously verify the integrity of containerized microservices and AI components as they migrate across the multi-provider edge-cloud continuum.

CL1.5 Automated Adversarial Testing & Continuous Purple Teaming: DevSecOps pipelines focus on code analysis and remediation; the adversarial counterpart provides continuous automated penetration testing, fuzzing (coverage-guided and protocol-level), and purple-team automation that simulates real attacker behaviour against live staging environments, ensuring defenses are validated continuously rather than at periodic intervals.

Cross-Layer 2: Dynamic Risk Management & Operational Cyber Resilience

CL2.1 Automated SOC Operations & Collaborative Threat Intelligence: Continuous monitoring, SOC federation across Member States, and automated incident containment, underpinned by legally protected, cross-border threat intelligence sharing ecosystems designed to overcome institutional trust barriers and incentivize active collaboration [5].

CL2.2 Systemic & Cascading Risk Mitigation: Mapping the evolving threat landscape and using impact-likelihood scoring to dynamically sever links and prevent cross-domain systemic failures.

CL2.3 High-Precision Telemetry and Security Signalling: Implementing “smart telemetry” and cybersecurity-related signalling to feed real-time threat intelligence into analytics platforms without overwhelming network bandwidth—ensuring that visibility does not itself become an attack vector or performance bottleneck.

CL2.4 Internet of Behavior (IoB) and Evidence Sharing: Utilizing cybersecurity and QoS signalling traffic to extract and share behavioral patterns across the infrastructure, establish-

ing an “Internet of Behavior” that enables proactive anomaly correlation and evidence-based risk assessment at continental scale.

CL2.5 AI-Native Analytics Integration (NWDAF & LLMSecOps): Routing cybersecurity signalling traffic and telemetry through centralized analytics hubs powered by AI agents. This creates a live “nervous system” for the network, allowing AI agents to continuously monitor for zero-day exploits, analyze encrypted traffic flows, and trigger automated incident responses aligned with NWDAF (Network Data Analytics Function) standards.

CL2.6 Cyber Threat Intelligence (CTI) Lifecycle Management: While CL2.1 and CL2.4 consume and act on threat data, explicit governance of the full CTI lifecycle—collection, processing, analysis, dissemination, and feedback—is required as a prerequisite for the federated SOC vision. A sovereign EU CTI platform must define this governance layer to ensure data quality, legal compliance, and attribution reliability.

CL2.7 Cyber Deception Technology & Active Defense (Honeynets, Canaries): Current detection is passive—telemetry is observed and anomalies flagged. Deception adds an active intelligence-collection dimension by deploying decoy assets (honeypots, honeynets, credential canaries, fake API endpoints) that generate high-fidelity, low-noise alerts when touched, enabling early detection of advanced intrusions and insider threats.

Cross-Layer 3: Interoperable Security Mechanisms & Open-Source Verifiability

CL3.1 Heterogeneous Interoperability & Simpl Integration: Secure communication and data exchange across disparate systems, edge nodes, and proprietary domains. This includes mandatory alignment with and integration into the Simpl open-source smart middleware [42] to ensure secure, federated access across Common European Data Spaces.

CL3.2 Minimal Interoperability Mechanisms (MIMs) & Open Verifiability: Establishing standardized, baseline security and privacy controls to ensure trust without creating vendor lock-in. This relies heavily on embracing open-source software and open hardware architectures as fundamental prerequisites for transparency, cryptographic verifiability, and the elimination of proprietary black-box vulnerabilities [42].

CL3.3 Neutral Host and Multi-Tenant Isolation: As the ecosystem shifts toward infrastructure sharing and Neutral Host models spanning wireless, optical, and cloud domains, robust isolation between tenants is critical. Security architectures must enforce Zero-Trust edge operations across different network partitions to prevent cross-tenant data leakage and privilege escalation.

CL3.4 Pan-European NaaS Federation and Cross-Border Edge Roaming: Establishing standardized interfaces and harmonized Network APIs (e.g., leveraging CAMARA) to interconnect the fragmented edge clouds of different European telecom operators into a unified Network-as-a-Service (NaaS) platform—enabling a “Schengen for Cloud/AI” where compute workloads can roam seamlessly across borders.

CL3.5 Semantic Security Interoperability (Common Ontologies & Data Models): Federated SOC and shared threat intelligence only function if participating organisations share a common semantic understanding of security events, asset taxonomies, and risk classifications. Standardized ontologies (e.g., aligned with STIX/TAXII and MITRE ATT&CK) are a prerequisite for meaningful cross-border correlation.

CL3.6 Cyber Range & Digital Twin Federation for Security Validation: Before deploying cross-border federated security configurations into production, they must be validated in high-fidelity simulation environments. ENISA already coordinates national cyber range inventories;

this cross-layer would federate them into a shared EU-wide infrastructure aligned with the Cyber Solidarity Act’s testing provisions.

Cross-Layer 4: Compliance-by-Design, Agile Governance & Automated Assurance

CL4.1 Policy-as-Code & Compliance-by-Design: Technical implementation of legal requirements (OSCAL) into machine-readable rules that pre-validate workloads.

CL4.2 Continuous Automated Auditing & Certification Lifecycle: Replacing “point-in-time” PDF certificates with real-time, telemetry-driven certification tracking for both cloud services (EUCS) [10] and physical ICT edge products (EUCC) [52]. This ensures that edge nodes autonomously maintain compliance by integrating mandatory vulnerability handling and cryptographic patch verification directly into their operational lifecycle, strictly aligning with EUCC maintenance and assurance rules [52].

CL4.3 Post-Quantum Readiness & Zero-Trust Architectures (ZTA): Moving beyond traditional perimeter-based defenses by embedding continuous, AI-driven Zero-Trust verification and micro-segmentation natively into the cloud-edge infrastructure, combined with a systematic migration roadmap to post-quantum cryptographic primitives across all layers.

CL4.4 Continuous Automated Certification: Replacing “point-in-time” PDF certificates with real-time, telemetry-driven certification tracking for both cloud services (EUCS) and physical ICT edge products (EUCC), ensuring that edge nodes autonomously maintain compliance by integrating mandatory vulnerability handling and cryptographic patch verification directly into their operational lifecycle [52].

CL4.5 Multicloud Key Management as a Service (KMaaS): When AI workloads, data flows, and context flows migrate dynamically across a multi-provider computing continuum, managing cryptographic keys and access policies becomes highly complex. KMaaS provides a provider-agnostic, sovereign key lifecycle management capability that prevents key sprawl and ensures revocation propagates consistently across heterogeneous environments.

CL4.6 Privacy-Preserving Federated Computation: When multiple providers work on the same infrastructure, AI training and inference may require access to datasets stored across different jurisdictions and competitive boundaries. Privacy-preserving techniques (federated learning, differential privacy, SMPC) enable collaborative computation while guaranteeing that raw data never leaves its sovereign jurisdiction.

CL4.7 Cyber Risk Quantification & Board-Level Reporting (CRQ): Compliance frameworks define what controls to implement. A dedicated cross-layer translates technical security posture into financial and operational risk language for decision-makers—potentially in real-time—enabling boards to make informed, evidence-based investment and risk-acceptance decisions aligned with DORA requirements.

CL4.8 AI Trustworthiness & Algorithmic Accountability: As AI systems themselves become actors within the security architecture—making autonomous containment decisions, classifying threats, orchestrating responses—the trustworthiness of those AI systems becomes a security-critical concern. This layer defines explainability, auditability, and override mechanisms for AI-driven security functions, ensuring alignment with the EU AI Act for high-risk systems.

CL4.9 Regulatory Digital Twins: Virtualized replicas of critical infrastructure specifically to stress-test regulatory compliance against evolving laws before applying changes to the live physical continuum.

7 Strategic Priority Clusters

8 Priority Cluster 1: Sovereign Trusted Infrastructure

Constituent Sub-Priorities: P1 (Hardware Roots of Trust & EUCC ‘High’), P2 (PQC Agility for Edge/IoT), P3 (Lightweight Cryptography & Physical-Layer Security), P8 (Confidential Computing & Privacy-Preserving AI), P9 (Resilient-by-Design & Island Mode).

Cluster Thesis: Before any federated or AI-driven security capability can be trusted, the substrate on which it executes must provide cryptographically verifiable integrity, quantum-resistant confidentiality, and autonomous survivability independent of external infrastructure. This cluster defines the verifiable hardware and cryptographic foundation that all higher-order capabilities in Clusters 2 and 3 depend upon as an architectural prerequisite.

7-Dimensional Analysis: Priority Cluster 1

1. Baseline & Target Objective

Baseline (2025–2026): European critical infrastructure and edge deployments operate under high foreign dependency across all five constituent sub-priorities. Hardware roots of trust are dominated by non-EU Trusted Platform Module (TPM) manufacturers — Infineon SLB9670, NXP SE050, and STMicroelectronics ST33 series collectively account for the majority of the EU market under ISO/IEC 11889 [54] and TCG TPM 2.0 specifications [55]. The Trusted Execution Environment (TEE) market is dominated by Intel TDX, AMD SEV-SNP, and ARM TrustZone, none of which are EU-designed or manufactured. Post-Quantum Cryptography (PQC) standardisation reached a definitive milestone with NIST’s publication of FIPS 203 (ML-KEM) [48], FIPS 204 (ML-DSA) [49], and FIPS 205 (SLH-DSA) [50] in August 2024; however, hardware-optimised implementations for 8/16-bit microcontrollers remain at the Applied R&D stage, as confirmed by NIST IR 8413 [45]. Physical-layer security for extreme-edge mMTC devices lacks standardised specifications. Edge nodes universally operate with hard cloud dependencies and no autonomous fallback architectures meeting the operational resilience mandates of NIS2 [7] and DORA [44].

Current cluster ODP: Applied R&D (driven by PQC, Lightweight PhySec, and Confidential Computing sovereign alternatives). Current cluster SSL: High Foreign Dependency.

Target (2040): At least 60% of EU critical infrastructure edge nodes deploy EU-designed RISC-V open-hardware secure elements certified at EUCC ‘High’ assurance level [52]. 100% of new EU critical IoT deployments support hybrid or full FIPS 203/204/205-compliant PQC. Zero-energy nodes execute quantum-resistant device authentication within a sub-100 μ J energy budget per operation. Trusted Execution Environments built on EU open-hardware architectures (Keystone [46], OpenTitan [56]) cover at least 50% of new EU cloud-edge deployments. All NIS2-essential and DORA-critical edge nodes demonstrate validated 72-hour autonomous Island Mode survivability independent of WAN connectivity.

2. Primary Drivers

Threat Drivers:

- State-sponsored APT exploitation of JTAG debug interfaces and electromagnetic side-channel analysis (EMSCA) on field-deployed edge nodes; the Spectre and Meltdown class of microarchitectural side-channel vulnerabilities establishes that even production-certified hardware can be compromised at scale, and far-edge nodes operating without physical security controls present a materially higher exposure surface than hyperscaler data centres.
- “Harvest Now, Decrypt Later” (HNDL) campaigns: state-level adversaries are actively archiving encrypted EU government and industrial communications under the expectation of Cryptographically Relevant Quantum Computer (CRQC) availability. NIST IR 8413 places the Q-Day horizon in the 10–20 year range with a highly uncertain lower bound [45]; HNDL renders this a present-tense threat against data with multi-decade confidentiality requirements.
- Physical tampering and environmental attacks on unattended far-edge nodes (smart grid substations, O-RAN radio units, industrial gateways) where the physical security controls available in hyperscaler facilities are absent.
- Cascading WAN disconnection failures propagating through centrally-dependent edge architectures, exploited by adversaries to degrade critical infrastructure services during hybrid conflict scenarios.

Regulatory Drivers:

- CRA Articles 10 and 11: mandatory vulnerability handling and security update obligations for products with digital elements; enforcement begins in 2027 for most product categories [8].
- NIS2 Article 21(2)(e): cryptographic policies and key management for essential and important entities; full enforcement began October 2024 [7].
- DORA Article 9: ICT resilience testing including hardware-level resilience requirements for financial sector critical infrastructure [44].
- AI Act Article 9: risk management system requirements for high-risk AI systems; critical infrastructure AI components require hardware integrity guarantees that only EUCC ‘High’ certified secure elements can provide [9].

3. High Impact Activities

- (a) **[Phase 1]** Develop sovereign European Secure Elements based on RISC-V open-hardware (building on OpenTitan [56] and Keystone [46] frameworks), targeting EUCC ‘High’ assurance level evaluation; implement automated hardware-level remote attestation protocols interoperable with TPM 2.0 [55] and ISO/IEC 11889 [54] for heterogeneous large-scale edge fleets.
- (b) **[Phase 1–2]** Optimise NIST FIPS 203 (ML-KEM) [48] and FIPS 204 (ML-DSA) [49] for 8/16-bit microcontrollers under severe energy and memory constraints (target: ML-KEM-512 within 32 KB RAM on Cortex-M0+ class devices); develop crypto-agility OTA update middleware enabling algorithm rotation without hardware replacement, compatible with ETSI TS 103 744 hybrid key exchange profiles [47].

- (c) **[Phase 1–2]** Implement ultra-lightweight PQC and keyless Physical-Layer Security (RF fingerprinting exploiting Channel State Information and Carrier Frequency Offset, CSI/CFO-based device authentication) for mMTC devices operating under zero-energy or near-zero-energy constraints; target integration with 6G ISAC (Integrated Sensing and Communication) interfaces per SNS JU SRIA [53].
- (d) **[Phase 2]** Deploy multi-party confidential computing pipelines using TEE abstraction layers (Keystone enclaves on RISC-V [46]) combined with Fully Homomorphic Encryption (FHE) schemes for collaborative AI training across multi-provider nodes without raw data exposure; align with EUCS ‘High+’ certification requirements [10].
- (e) **[Phase 2–3]** Engineer autonomous Island Mode architecture: soft-state edge orchestration using cached credentials with 72-hour validity windows, LoRaWAN LPWAN fallback micro-slices, eBPF-based kernel-level policy enforcement for autonomous node isolation, and bio-inspired self-healing swarm protocols enabling edge nodes to sustain critical service delivery without cloud connectivity or human intervention.

4. Catalysts & Enablers

Funding Alignment: Horizon Europe Cluster 4 (Digital, Industry and Space) for sovereign secure silicon and PQC hardware implementation; Horizon Europe Cluster 3 (Civil Security for Society) for edge resilience and Island Mode architectures; EU Chips Act (Regulation (EU) 2023/1781) pilot fabrication lines for security-critical silicon via the Chips Joint Undertaking (KDT JU); ECCC Strategic Agenda priority areas “Secure Components and Systems” and “Post-Quantum Cryptography transition infrastructure” [41]; Digital Europe Programme for PQC migration tooling and crypto-agility middleware.

Standardisation: NIST NCCoE “Migration to Post-Quantum Cryptography” project (reference architecture and migration playbooks); CEN/CENELEC JTC 13/WG 8 (hardware security standards aligned with EUCS Common Criteria); ETSI TC CYBER (ETSI TS 103 744 quantum-safe hybrid key exchanges [47]); ENISA PQC transition guidelines and cryptographic inventory tooling; 3GPP SA3 for physical-layer security integration with 5G/6G specifications [57].

Ecosystem Enablers: RISC-V International open-hardware community; OpenTitan coalition pilot deployments for EU sovereign RoT [56]; SNS JU pilot networks for 6G-integrated PhySec validation [53]; federated Cyber Range infrastructure (CL3.5) for Island Mode and TEE attestation stress-testing.

5. Disruptors & Systemic Risks

- **Q-Day cliff:** CRQC becoming operational before EU PQC migration completes; NIST IR 8413 places this horizon at 10–20 years with a highly uncertain lower bound [45]. The EU PQC migration timeline targeting critical infrastructure completion by 2030 has limited buffer against the pessimistic lower bound, and HNDL campaigns make this a present-tense threat against long-lived secrets.
- **EU Chips Act EUV lithography shortfall:** RISC-V secure element production at the node geometries required for EUCS ‘High’ certification depends on advanced lithography; EU 2030 Chips Act targets (20% global market share) do not specify security-critical silicon capacity independently, and ASML EUV tooling remains subject to US–

Netherlands export control dynamics that could constrain sovereign fabrication timelines beyond 2028.

- **Side-channel analysis breakthroughs against FIPS 203 software implementations:** lattice-based algorithms have demonstrated timing and cache-timing attack surfaces in software implementations on constrained devices; hardware countermeasures are not yet standardised at the time of writing [CITATION REQUIRES VERIFICATION — recent SCA against ML-KEM implementations].
- **TEE architectural monoculture:** if EU open-hardware TEE adoption concentrates on a single architecture (e.g., Keystone only [46]), a single discovered vulnerability propagates across the entire EU critical infrastructure TEE estate without architectural diversity as a containment layer.
- **LPWAN regulatory fragmentation:** Island Mode fallback using LoRaWAN operates on unlicensed 868 MHz spectrum in the EU; spectrum policy fragmentation across Member States could prevent coordinated emergency micro-slice activation under unified crisis response scenarios.

6. Failsafe Mechanisms & Resilience

- **Hybrid cryptographic transition:** deploy ML-KEM (FIPS 203) [48] combined with ECDH-P384 in a hybrid key encapsulation scheme per ETSI TS 103 744 [47]; if ML-KEM is found vulnerable via cryptanalytic breakthrough, ECDH-P384 maintains classical confidentiality while SLH-DSA (FIPS 205) [50] provides backup hash-based signature integrity with distinct mathematical foundations.
- **TEE implementation diversity:** deploy a minimum of two architecturally distinct TEE technologies across critical EU infrastructure (ARM CCA on server-class hardware and RISC-V Keystone [46] on edge nodes) to prevent single-architecture compromise from cascading across the entire confidential computing estate.
- **Autonomous node quarantine via eBPF:** if TEE remote attestation fails, eBPF-based network policy enforcement at the Linux kernel level (e.g., via Cilium or Falco eBPF rulesets) isolates the affected node from the orchestration plane within sub-second latency, blocking lateral movement before human intervention is available.
- **Island Mode failsafe:** edge nodes maintain cryptographically signed cached credentials with defined 72-hour validity windows; on WAN disconnection, a LoRaWAN LPWAN micro-slice activates a minimal viable network carrying life-safety telemetry; if LPWAN also fails, nodes revert to local-only operation with integrity-protected audit logging for post-incident forensics.
- **RF fingerprinting fallback:** for zero-energy IoT nodes where cryptographic processing exceeds the available energy budget, Physical-Layer authentication via RF fingerprinting (CSI/CFO-based device identity) provides keyless device verification using intrinsic wireless channel characteristics, requiring no cryptographic key storage or computation.

7. Transversal Meta-Layer

Sovereignty & Autonomy: This cluster eliminates approximately 100% foreign dependency on non-EU TPMs (Infineon, NXP, STMicroelectronics dominate the EU market under TCG

TPM 2.0 [55]) and non-EU TEE architectures (Intel TDX, AMD SEV-SNP, ARM TrustZone, all non-EU designed and manufactured). Crypto-agility frameworks additionally reduce dependency on US-controlled NIST algorithm supply chains by enabling algorithm substitution if standardisation processes are constrained by geopolitical factors.

Trust & Ethics: GDPR Article 25 (data protection by design and by default) is operationalised through TEE-enforced data minimisation and in-use encryption in multi-tenant environments, ensuring that sensitive data processing cannot be accessed by platform operators [51]. AI Act Article 9 risk management system requirements for high-risk AI components require hardware integrity guarantees — only EUCC ‘High’ certified secure elements provide the mathematically verifiable trust anchor this obligation demands [9].

Sustainable Security (Green IT): CRYSTALS-Kyber (ML-KEM) demonstrates approximately $1.5\times$ computational overhead relative to ECDH at equivalent security levels on ARM Cortex-M4, per NIST IR 8413 [45], representing a manageable energy penalty. Lightweight PQC research targets sub-100 μJ per authentication operation for zero-energy IoT nodes. EU-sovereign RISC-V secure elements designed for edge power constraints enable more energy-efficient cryptographic operations than general-purpose processor TPM emulation, reducing per-device carbon footprint at continental deployment scale.

8.1 Progression Matrix: Priority Cluster 1 — Sovereign Trusted Infrastructure

Continued...

SSL	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Sovereignty
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Table 3: **Cluster 1: Sovereign Trusted Infrastructure** (P1, P2, P3, P8, P9). Transition matrix across the four methodology dimensions. Sub-priority labels in parentheses indicate where individual priorities diverge from the cluster-level descriptor within a cell. ODP scale: Applied R&D → Fragmented Piloting → Standardised Integration → Ubiquitous Operation. SSL scale: High Foreign Dependency → Emerging EU Ecosystem → Full Sovereign Autonomy.

SSL	R&I Maturity ODP indicator per sub-priority	Policy, Regulatory & Standards enforcement · certification · standardisation	Operational Deployment & Adoption actor types · scale · sector coverage	Market Uptake & Strategic Sovereignty SSL indicator · dependency reduction · industrial capacity
& Foreign y	Cluster ODP: Applied R&D / Fragmented Piloting The cluster is split at baseline. (<i>P1, P9</i>) reach Fragmented Piloting; RISC-V secure element projects targeting EUCC ‘High’ are funded under Horizon Europe and EU Chips Act; Island Mode soft-state protocols are validated in 5G research testbeds. (<i>P2, P3, P8</i>) remain at Applied R&D: NIST FIPS 203/204/205 algorithms are being optimised for 8/16-bit microcontrollers under severe energy and memory constraints; Keystone and OpenTitan prototypes are under EUCC pre-evaluation; RF fingerprinting Physical-Layer Security is validated in controlled 5G lab environments; Fully Homomorphic Encryption approaches practical latency bounds in research settings only. <i>The cluster ODP is Applied R&D, driven by the majority of sub-priorities. Deployment claims for P2, P3, P8 must be grounded in named Horizon Europe or EUCC project outcomes, not technology previews.</i>	Regulatory environment: enforcing, certification nascent NIS2 Article 21(2)(e) cryptographic policies for essential entities entered full enforcement in October 2024. CRA Articles 10 and 11 (vulnerability handling and security update obligations) begin enforcement in 2027, creating immediate procurement-level demand for hardware security certification. AI Act Articles 9 and 15 establish hardware integrity requirements for high-risk AI systems and enter applicability in 2025–2026. ENISA PQC migration guidelines are published; EUCC evaluation processes are initiated for the first RISC-V prototypes, but no EU-specific EUCC ‘High’ certified sovereign secure element exists yet. CEN/CENELEC JTC 13 standards for hardware security MIMs are in draft phase. <i>The regulatory framework is enforcing existing obligations but lacks EU-specific hardware certification outputs; the critical gap is the absence of a ratified EUCC ‘High’ EU-origin product.</i>	Deployment: research consortia and national agencies only RISC-V secure element pilots are operated by 3–5 Horizon Europe consortia and national cybersecurity agencies in controlled environments. Intel TDX, AMD SEV-SNP, and ARM TrustZone dominate all operational TEE deployments across EU cloud and critical infrastructure. PQC is confined to cloud-side software implementations in pilot contexts; far-edge IoT devices run classical cryptography exclusively. Edge nodes operate with hard cloud dependencies and no validated autonomous fallback architecture meeting DORA or NIS2 business continuity thresholds. SMEs have no access to EU-sovereign hardware alternatives at viable cost. <i>Operational deployment at Phase 1 is institutional and research-concentrated. The gap between pilot existence and operational availability for regulated-sector procurement must be explicitly acknowledged.</i>	SSL: High Foreign Dependency Non-EU TPMs (Infineon SLB9670, NXP SE050, STMicroelectronics ST33) hold ≥90% of EU market share in hardware security modules. Intel, AMD, and ARM control the entire operational TEE market. No EU-designed and EU-manufactured secure element is commercially available. EU Chips Act fabrication investment for security-critical silicon is initiated but pre-production. US NIST algorithm supply chain is the sole PQC standardisation reference. Structural vulnerability is direct: geopolitical restriction of access to any of the three major TPM suppliers would immediately impair EU critical infrastructure attestation capability. <i>Quantified dependency at Phase 1 baseline: ≥90% foreign market share, zero EU-certified sovereign hardware alternatives. Sovereignty progress is measured from this specific figure.</i>

Continued. . .

SL	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Sovereignty
	R&I trigger: At least one EU-designed RISC-V secure element prototype has been independently validated against EUCC ‘High’ evaluation criteria and has entered the formal ENISA/national scheme certification process. Policy trigger: CRA Articles 10 and 11 are in active enforcement with documented non-compliance decisions, confirming that market incentives for certified EU hardware alternatives are financially material. Operational trigger: Documented operational deployment of hybrid ML-KEM (FIPS 203) plus ECDH-P384 PQC schemes in at least one EU critical infrastructure sector (energy or financial) at production scale, not pilot scale. Market trigger: EU Chips Act security-critical silicon fabrication pilot line has produced and delivered its first commercially viable batch of EU-origin secure elements, confirmed by independent yield and cost assessment.			
Industrial wide 3 tion gging tem	Cluster ODP: Standardised Integration / continued Applied R&D (P1, P9) reach Standardised Integration: the first EUCC ‘High’ certified EU-manufactured RISC-V secure elements are in production and being integrated into critical infrastructure; automated remote attestation at fleet scale is validated; 72-hour Island Mode achieves standardised certification under 3GPP/ETSI fallback API profiles. (P2, P3) transition from Applied R&D to Fragmented Piloting: hybrid ML-KEM plus ECDH-P384 schemes are validated at Industrial IoT scale; lightweight PQC for sub-100 μJ zero-energy nodes is demonstrated under 6G ISAC testbeds. (P8) reaches Fragmented Piloting: RISC-V Keystone TEEs are operational in multi-provider Data Spaces; FHE performance approaches the threshold for selective production use in high-value financial analytics. Phase 2 R&I is not purely consolidation: three of five sub-priorities are still in active applied and industrial R&D. Describing this phase as “standardisation only” misrepresents the maturation arc.	EUCC certification and PQC standards formalised The EUCC ‘High’ certification of the first EU-manufactured RISC-V secure element is the definitive Phase 2 policy gate: it transforms applied research into a market-accessible, legally certifiable product and unlocks the procurement pathway for EU critical infrastructure operators. NIST FIPS 203/204/205 are adopted into ETSI TS profiles for EU deployment contexts. ENISA publishes a mandatory PQC migration timeline for critical infrastructure operators, referenced in NIS2 sector-specific implementing acts. EU Chips Act fabrication investments are active and measurable. CEN/CENELEC hardware security standards for EU-origin secure elements enter ratification. The single Phase 2 policy gate is EUCC ‘High’ certification of a EU-origin product. All other policy milestones are enabling conditions, not gate conditions.	Critical infrastructure and regulated sectors deploying EU alternatives EU EUCC ‘High’ certified secure elements are deployed in energy sector substations and telecommunications critical infrastructure in at least five Member States. Hybrid PQC schemes are standard in new Industrial IoT and financial sector (DORA-compliant) deployments. RISC-V Keystone TEEs operate in federated multi-provider Data Spaces alongside existing Intel TDX infrastructure. Validated 72-hour Island Mode is deployed in at least three EU 5G cross-border corridor pilots under SNS JU governance. SME-accessible PQC tooling and SBOM compliance support are available commercially. Phase 2 deployment evidence requires named sector and Member State scope; anonymous “multi-sector deployment” claims do not constitute verifiable operational maturity.	SSL: Emerging EU Ecosystem EU-designed secure elements achieve 15–30% market share in new critical infrastructure procurement, rising from a near-zero baseline. EU Chips Act fabrication investments have created a domestic production capacity that is commercially credible, if not yet dominant. Foreign TPMs remain the standard for legacy deployments but are no longer the only option for new regulated-sector procurement. EU open-hardware TEE ecosystem (Keystone, OpenTitan) holds commercially certifiable products. The conditions for Full Sovereign Autonomy are structurally in place — fabrication capacity, certification infrastructure, standards leadership — but have not yet reached continental operational scale. SSL transition requires industrial substance, not research demonstration. 15–30% new critical infra market share is the quantified Phase 2 SSL threshold.

Continued. . .

SL	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Sovereignty
	R&I trigger: Production-ready PQC-native RISC-V silicon is available from EU-domiciled fabrication at commercially viable yields, confirmed by independent cost-per-unit assessment against equivalent non-EU alternatives. Policy trigger: EUCC ‘High’ certification is mandated as a procurement requirement for all new EU critical infrastructure edge node deployments, with no waiver provisions for new procurements (legacy replacement programmes are time-bound). Operational trigger: Validated 72-hour Island Mode is deployed as a standard operational baseline — not a pilot — in ≥50% of new EU NIS2-essential entity edge node procurements in the energy and telecommunications sectors. Market trigger: EU Chips Act sovereign fabrication capacity achieves ≥50% self-sufficiency for security-critical silicon in new EU critical infrastructure procurement, measured by share of new orders fulfilled by EU-origin fabrication.			
	Cluster ODP: Ubiquitous Operation All five sub-priorities operate at Ubiquitous Operation. EU-designed RISC-V secure elements are the hardware default across all new EU ICT products. Fully Homomorphic Encryption is operationalised for end-to-end confidential processing from IoT to HPC, not confined to high-value analytics. Zero-energy PhySec authentication via RF fingerprinting operates as a standard device identity mechanism for battery-less mMTC nodes. Bio-inspired swarm protocols enable autonomous cyber-physical recovery without cloud intervention. Phase 3 R&I is oriented toward the next-generation frontier: neuromorphic secure compute, post-second-generation quantum hardware, and post-CMOS security architectures — EU research groups define the agenda rather than adopting it. <i>Ubiquitous Operation requires deployment statistics, not deployment plans. Phase 3 evidence must cite procurement share data and operational audit findings, not programme milestone completions.</i>	EU as international hardware security standards reference EUCC ‘High’ certification is mandatory for all new EU ICT products without waiver provisions. PQC-native silicon is the EU procurement default. EU hardware security standards are referenced in ISO/IEC international standardisation and in bilateral digital trade agreements. Continuous automated EUCC lifecycle maintenance replaces point-in-time audit models completely. The EU sets the terms of hardware security compliance for market access, transforming its internal market standard into a de facto international reference. <i>Phase 3 policy maturity requires evidence of international standards influence: at least one EU hardware security standard adopted or formally referenced in an ISO/IEC or ITU-T international standard.</i>	Security-by-Design the unconditional hardware and crypto default ≥60% of EU critical infrastructure edge nodes deploy EUCC ‘High’ certified EU-designed secure elements. FHE is operational end-to-end from IoT sensor to HPC compute node. Autonomous Island Mode provides validated 72-hour survivability as a mandatory operational baseline across all NIS2-essential and DORA-critical edge deployments. No new EU ICT product enters the market without a hardware root of trust. The question “does this deployment have hardware-level integrity?” has no non-trivial answer for any new EU deployment. <i>Phase 3 deployment normalisation: ≥60% market penetration for EU secure elements in critical infra; 100% of new EU ICT products with hardware RoT as a default.</i>	SSL: Full Sovereign Autonomy EU-designed and EU-manufactured secure elements hold ≥60% of new critical infrastructure procurement market share. Foreign legacy dependency operates under binding replacement programmes with defined timelines. EU sovereign RISC-V silicon and open-hardware TEE architectures are exported to and adopted by partner jurisdictions. EU PQC and hardware TEE standards are the international reference. Structural exposure to geopolitical disruption of any single non-EU hardware supplier has been eliminated for new deployments. The EU has moved from defensive sovereignty (reducing dependency) to offensive strategic posture (defining the terms of global hardware security competition). <i>Full Sovereign Autonomy is verified by ≥60% market share in new critical infra procurement plus the existence of binding replacement programmes for legacy foreign hardware — not by 100% market share.</i>

9 Priority Cluster 2: Cognitive Security and Autonomous Cyber Defence

Constituent Sub-Priorities: P6 (Cognitive SecOps / LLMSecOps & Automated Threat Intelligence), P10 (Proactive APT Management), and the AI-as-target dimension drawn from CL1.2 (Model Integrity / DataBOMs), CL1.3 (Dynamic Supply Chain Verification / Runtime SBOMs), and CL4.7 (AI Trustworthiness & Algorithmic Accountability).

Cluster Thesis: Artificial intelligence is simultaneously the most powerful instrument available for autonomous cyber defence and the most consequential new attack surface introduced into the security architecture. A roadmap that deploys AI agents for autonomous threat containment without hardening those agents against adversarial manipulation creates a critical second-order vulnerability. This cluster therefore addresses three inseparable sub-dimensions: AI as security instrument (LLMSecOps and APT hunting), AI as adversary amplifier (AI-generated APT tooling), and AI as target (model poisoning, adversarial evasion, and algorithmic accountability for autonomous containment decisions).

7-Dimensional Analysis: Priority Cluster 2

1. Baseline & Target Objective

Baseline (2025–2026): European Security Operations Centres (SOCs) rely predominantly on rule-based SIEM platforms — Splunk (US-domiciled, acquired by Cisco in 2023) and IBM QRadar dominate enterprise deployments. AI-based tools are deployed as analyst “copilots” with human analysts retaining all containment authority; no EU-sovereign foundation models for security operations exist at production scale. APT dwell times in European enterprise environments average 150–200 days, with state-sponsored threat actors demonstrating extended persistence in EU critical infrastructure. The adversarial robustness of AI security agents — their resistance to prompt injection, model evasion, and training data poisoning — is addressed only in research contexts without operational certification frameworks. NWDAF integration per 3GPP TS 23.288 [58] exists in 5G standalone deployments but remains fragmented across operators, and no EU-mandated framework for the trustworthiness of autonomous security AI exists prior to AI Act enforcement.

Current cluster ODP: Applied R&D (AI-as-target and sovereign LLM development); Siloed AI Copilots (P6); Multi-Layered Alert Correlation (P10). Cluster ODP: Applied R&D. Cluster SSL: High Reliance on US Foundation Models / High Dependency on Global Threat Feeds.

Target (2040):

Swarm-based agentic AI defends the EU Cloud-Edge Continuum autonomously, with APT dwell time reduced to under 24 hours in certified critical infrastructure deployments. EU-sovereign edge Small Language Models (SLMs) and Large Context Models (LCMs) for security operations are certified under AI Act Article 9 and cover at least 50% of EU critical infrastructure SOC deployments. All autonomous security agents pass mandatory adversarial robustness validation before operational deployment. Moving Target Defense reduces APT reconnaissance success rates below 5% in validated simulation environments. EU AI security frameworks are adopted as global reference implementations by partner jurisdictions.

2. Primary Drivers

Threat Drivers:

- Nation-state APT campaigns (attributed by EU agencies to APT28/GRU, APT29/SVR, and Sandworm) increasingly exploit LLM hallucination and prompt injection vulnerabilities to manipulate autonomous security agents into executing incorrect containment actions — weaponising the defence architecture against itself.
- AI-generated spear-phishing and polymorphic malware enabling automated, highly-targeted APT campaigns that adapt faster than signature-based detection cycles, invalidating the static detection rules that rule-based SIEM platforms depend upon.
- Supply chain attacks targeting AI model training pipelines: data poisoning and back-door insertion into security AI models deployed at continental scale can silently degrade detection capability without triggering observable anomalies in standard monitoring.
- Systematic abuse of MITRE ATT&CK-catalogued techniques (T1027 Obfuscated Files or Information; T1055 Process Injection) within AI-assisted reconnaissance tools, enabling more efficient bypass of detection rules derived from those same frameworks.

Regulatory Drivers:

- AI Act Article 9: risk management system requirements for high-risk AI systems — autonomous security containment agents making access revocation or network isolation decisions qualify as high-risk systems requiring mandatory conformity assessment before operational deployment [9].
- AI Act Article 15: accuracy, robustness, and cybersecurity requirements for high-risk AI, including specific resilience requirements against adversarial manipulation of inputs and training data [9].
- AI Act Article 13: transparency and provision of information for high-risk AI — autonomous security decisions must be explainable to affected entities and competent national authorities [9].
- NIS2 Article 21(2)(b): incident handling procedures, including AI-augmented detection and response, for essential entities [7].

3. High Impact Activities

- (a) **[Phase 1]** Deploy NWDAF-integrated (3GPP TS 23.288 [58]) telemetry pipelines feeding ML-based UEBA (User and Entity Behaviour Analytics) for APT indicator correlation; establish behavioural baseline profiles across federated enterprise environments targeting reduction of APT dwell time from the current 150–200 day average to under 72 hours in pilot deployments via ML-assisted Indicator of Compromise (IoC) pattern detection.
- (b) **[Phase 1–2]** Develop EU-sovereign Small Language Models (SLMs, under 10B parameters) and Large Context Models (LCMs) for edge-native SOC operations (LLMSec-Ops), trained on MITRE ATT&CK-aligned threat intelligence corpora under GDPR-compliant data governance [51]; establish mandatory AI Act Article 9 conformity assessment pipelines for all autonomous security agent deployments prior to production authorisation [9].

- (c) **[Phase 2]** Deploy edge-native “Guardian” nodes executing autonomous Zero-Trust re-configurations per NIST SP 800-207 ZTA principles [59] and Moving Target Defense (MTD) — dynamically shuffling IP addressing, network topology, and software diversity to disrupt APT reconnaissance — governed by ETSI GS ZSM 009 closed-loop automation standards [60].
- (d) **[Phase 2]** Establish mandatory adversarial robustness validation pipelines for all AI security agents prior to operational deployment: automated red-team adversarial testing (prompt injection, model evasion, backdoor detection via CL1.4 purple-team automation), with continuous certification under a CEN/CENELEC-standardised AI security agent assurance framework aligned with CL4.7 AI Trustworthiness requirements.
- (e) **[Phase 3]** Deploy swarm-based agentic AI coordination protocols enabling decentralised, self-evolving threat intelligence synthesis across the EU Cloud-Edge Continuum; integrate with Deception Technology infrastructure (CL2.6: honeypots, honeynets, credential canaries, fake API endpoints) to generate high-fidelity, low-noise adversarial ground truth for continuous autonomous agent retraining.

4. Catalysts & Enablers

Funding Alignment: EuroHPC Joint Undertaking AI Factories for training EU-sovereign security foundation models at the compute scale required; ECCC Strategic Agenda priority areas “Cybersecurity of AI” and “AI-enabled cybersecurity tools and services” [41]; Horizon Europe Cluster 3 (Civil Security for Society) calls targeting AI-driven threat intelligence and autonomous incident response; EU AI Office for conformity assessment infrastructure supporting high-risk security AI certification under the AI Act.

Standardisation: ETSI GS ZSM 009 (Zero-Touch Network and Service Management, Closed-Loop Automation [60]); 3GPP TS 23.288 (NWDAF architecture for AI-native 5G network analytics [58]); MITRE ATT&CK framework (behavioural threat intelligence taxonomy for agent training corpora); NIST SP 800-207 (Zero Trust Architecture [59]); CEN/CENELEC JTC 13 working groups on AI security agent assurance standardisation.

Ecosystem Enablers: ENISA AI cybersecurity threat landscape guidelines; federated Cyber Range infrastructure (CL3.5) for adversarial agent testing under realistic attack scenarios; CONCORDIA successor programme for EU-wide SOC federation and cross-border CTI sharing [5]; EU AI Act notified conformity assessment bodies for high-risk AI product certification.

5. Disruptors & Systemic Risks

- **Adversarial AI arms race:** threat actors weaponising LLMs for automated zero-day discovery and polymorphic exploit generation may outpace the defensive AI maturation timeline — specifically, AI-generated vulnerability discovery could exceed human patching bandwidth within the Phase 1–2 window, before EU-sovereign autonomous response agents are certified for operational deployment.
- **LLM hallucination in containment agents:** false positive isolation of critical infrastructure nodes (e.g., a hospital network segment incorrectly quarantined during a medical emergency) constitutes an operational safety risk that may impose worse harm than the threat the agent was designed to contain — this is the central argument for mandatory HITL escalation in Phase 1.

- **EU sovereign LLM compute deficit:** training security-relevant foundation models requires EuroHPC-scale compute; as of 2025–2026, EU frontier model training capacity lags US and China by approximately one order of magnitude, constraining Phase 1 EU-sovereign models to SLM/LCM scale (<10B parameters) and limiting capability relative to US foundation models.
- **AI model poisoning via CTI feed corruption:** if adversaries inject false indicators into NWDAF analytics pipelines or Cyber Threat Intelligence (CTI) sharing platforms (CL2.1), behavioural baselines trained on poisoned data will silently misclassify legitimate traffic and fail to detect real APTs — a harder failure mode to detect than outright system unavailability.
- **AI Act certification bottleneck:** mandatory conformity assessment for high-risk autonomous security agents may create a 24–36 month regulatory gap during which EU organisations operate without certified autonomous containment, relying on slower human-in-the-loop processes that cannot match AI-assisted adversary speed.

6. Failsafe Mechanisms & Resilience

- **Shadow mode operation:** all autonomous AI security agents run in parallel with human-supervised rule-based SIEM during Phases 1–2; containment actions affecting Tier-1 critical infrastructure (energy, healthcare, transport per NIS2 Annex I) require mandatory Human-In-The-Loop (HITL) escalation regardless of AI confidence score, with reversible quarantine actions bounded by a defined 60-minute rollback window [7].
- **Model diversity and behavioural circuit breakers:** deploy a minimum of two architecturally distinct AI detection models (e.g., transformer-based LLM plus graph neural network UEBA) to prevent a single-model adversarial exploit from disabling the entire detection capability; behavioural circuit breakers automatically suspend any agent exhibiting an anomalous action rate or pattern deviating beyond 3σ from its established operational baseline, reverting to rule-based SIEM until re-certification.
- **RLHF continuous correction:** incorporate analyst override decisions into continuous Reinforcement Learning from Human Feedback (RLHF) retraining to prevent model drift from training distribution; quarterly adversarial red-team audits (CL1.4) detect capability degradation before it reaches operational significance.
- **Air-gapped CTI validation:** all external CTI feeds ingested by NWDAF analytics pipelines pass through an isolated validation environment where indicator quality is assessed against independent behavioural ground truth before poisoned data can influence operational AI model weights.
- **Deception-backed calibration:** honeynet and credential canary infrastructure (CL2.6) provides high-fidelity, low-noise ground truth against which AI behavioural models are continuously calibrated, reducing false positive rates and providing an independent signal for detecting model poisoning via discrepancy analysis between canary alerts and AI model predictions.

7. Transversal Meta-Layer

Sovereignty & Autonomy: This cluster eliminates reliance on US-domiciled SIEM platforms (Splunk/Cisco, IBM QRadar) and US foundation models for operational security intelligence.

EU-sovereign SLMs trained under GDPR-compliant data governance [51] ensure that sensitive threat intelligence and behavioural telemetry does not transit non-EU cloud infrastructure. EU leadership in trustworthy AI security tools creates an exportable technology capability aligned with the AI Act’s ambition for global regulatory standard-setting.

Trust & Ethics: AI Act Article 13 (transparency requirements) is mandated for all autonomous security decisions — affected entities must receive meaningful explanations for AI-driven access revocations or network isolation actions [9]. GDPR Article 22 (restrictions on automated individual decision-making) applies where AI containment decisions affect individual data subjects; human review mechanisms must be structurally embedded, not optional add-ons [51]. AI Act Article 15 adversarial robustness requirements ensure security AI systems cannot be weaponised against the infrastructure they are designed to protect [9].

Sustainable Security (Green IT): Edge-native SLMs under 10B parameters reduce inference energy consumption by two to three orders of magnitude compared to centralised GPT-class model inference, enabling real-time threat classification at the point of detection without data centre energy overhead. Federated training paradigms reduce repeated data transmission overhead compared to centralised training architectures and are compatible with the privacy preservation requirements of GDPR Article 25 [51].

9.1 Progression Matrix: Priority Cluster 2 — Cognitive Security and Autonomous Cyber Defence

Continued...

SSL	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Sovereignty
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Table 4: **Cluster 2: Cognitive Security & Autonomous Cyber Defence** (P6, P10, AI-as-target dimension from CL1.2, CL1.3, CL4.7). Transition matrix across the four methodology dimensions. The AI-as-target dimension (model integrity, adversarial robustness, algorithmic accountability) is labelled (*AI-T*) where it diverges from P6 or P10 descriptors.

SSL	R&I MaturityODP indicator per sub-priority	Policy, Regulatory & Standardsenforcement · certification · standardisation	Operational Deployment & Adoptionactor types · scale · sector coverage	Market Uptake & Strategic SovereigntySSL indicator · dependency reduction · industrial capacity
& n tion	Cluster ODP: Applied R&D / Siloed AI Copilots (P6) reaches Siloed AI Copilots: commercial LLMs (GPT-4-class) assist human SOC analysts in alert triage, threat report generation, and MITRE ATT&CK technique mapping; no EU-sovereign LLM for security operations exists at production scale; NWDAF (3GPP TS 23.288) telemetry pipelines are integrated in 5G SA pilot deployments for anomaly detection. (P10) is at Multi-Layered Alert Correlation: ML-based UEBA is applied to APT IoC detection in research and pilot contexts; APT dwell time reduction from the 150–200 day average is demonstrated in controlled settings. (AI-T) is at Applied R&D: adversarial robustness testing (prompt injection, model evasion, DataBOM verification) is an academic research function without operational certification frameworks. Phase 1 R&I distinguishes three sub-dimensions with different maturity levels. Conflating them into a single ODP cell obscures that AI-as-target capability is the least mature and the most consequential gap.	AI Act enforcing; conformity assessment infrastructure absent AI Act Articles 9 (risk management), 13 (transparency), and 15 (adversarial robustness) enter applicability in 2025–2026, establishing mandatory requirements for autonomous security AI as high-risk systems. However, no EU conformity assessment bodies with capacity to certify autonomous security agents are operational at the start of Phase 1. ETSI GS ZSM 009 (closed-loop automation) is published but not mandated. NIS2 Article 21(2)(b) incident handling procedures are being operationalised across essential entities. ENISA AI cybersecurity threat landscape guidelines are published. The critical regulatory gap is the absence of a standardised adversarial robustness test suite applicable to security-specific AI agents. The AI Act creates obligations before the conformity assessment infrastructure exists to fulfil them. This 24–36 month gap is the dominant Phase 1 policy risk for this cluster.	Deployment: SOC copilot use by large enterprises and national CSIRTs US-platform LLMs (GPT-4-class, under data processor agreements) operate as analyst copilots in large enterprise and national CSIRT SOC. Rule-based SIEMs (Splunk/Cisco, IBM QRadar) remain the primary detection infrastructure across EU regulated sectors. NWDAF-integrated AI analytics operate in 3–5 5G SA deployments run by major EU telecom operators. No certified autonomous security containment agent is in operational deployment anywhere in the EU. APT dwell times of 150–200 days in EU enterprise environments reflect the gap between AI-assisted detection capability and operational deployment at scale. SMEs lack access to AI-powered SOC tooling at viable cost. Phase 1 deployment is concentrated in the largest institutional actors (national CSIRTs, Tier-1 operators, major financial entities). The gap between these deployments and SME access defines the adoption challenge for Phase 2.	SSL: High Reliance on US Foundation Models The EU SOC tooling market is dominated by US-domiciled platforms: Splunk (Cisco), IBM QRadar, and Microsoft Sentinel collectively control the EU enterprise SIEM market. The foundation models used for SOC copilot functions are US-origin (OpenAI, Anthropic, Google) and are accessed under data processor agreements, creating persistent extra-territorial data exposure. No EU-sovereign foundation model at production scale for security operations exists. EuroHPC compute capacity is insufficient for frontier-class security model training. The adversarial robustness validation market is pre-commercial. Quantified dependency: US platforms hold ≥80% of EU enterprise SIEM market; US-origin models are the sole source for LLM-based SOC tooling. This specific dependency structure is what Phases 2 and 3 must reduce.

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SL	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Sovereignty
	R&I trigger: At least one EU-sovereign Small Language Model (≤ 10 B parameters) trained on GDPR-compliant security corpora has been validated against an independently benchmarked EU critical infrastructure SOC use case at production-equivalent scale. Policy trigger: AI Act conformity assessment bodies with capacity to certify autonomous security AI as high-risk systems under Articles 9 and 15 are operational and have issued at least one certificate to a security AI agent product. Operational trigger: At least one EU-sovereign LLMSecOps agent is in certified operational deployment in a named NIS2-essential entity, demonstrating APT dwell time below 72 hours in a documented production environment. Market trigger: A standardised adversarial robustness test suite for security AI agents, developed under CEN/CENELEC or ETSI TC CYBER governance, has been published and is referenced in at least one EU procurement framework for critical infrastructure AI tools.			

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SL	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Sovereignty
Industrial side y tion eign erging	Cluster ODP: Federated Edge Agents / Automated Triage (P6) reaches Federated Edge Agents: EU-sovereign SLMs ($\leq 10B$ parameters) trained on MITRE ATT&CK-aligned corpora are operational at the edge for real-time SOC analytics without cloud inference dependency; Guardian nodes execute autonomous Zero-Trust reconfigurations per NIST SP 800-207. (P10) reaches Automated Triage and LLMSecOps Containment: LLMSecOps agents ingest CTI feeds, track APT lateral movement against behavioural frameworks, and orchestrate incident triage autonomously under ETSI GS ZSM 009 governance; APT dwell time ≤ 72 hours in certified pilot deployments. (AI-T) reaches Standardised Integration: adversarial robustness validation pipelines (prompt injection, model evasion, backdoor detection) are standardised and mandatory under AI Act Article 9 conformity assessment; continuous RLHF retraining and circuit-breaker mechanisms are in production. Phase 2 R&I includes active applied and industrial R&D for all three sub-dimensions — swarm AI protocols, post-quantum AI security, and next-generation adversarial testing frameworks — not merely consolidation of Phase 1 results.	AI Act conformity assessment operational: the Phase 2 gate The single Phase 2 policy gate is the operationalisation of a standardised AI Act conformity assessment framework for autonomous security agents, enabling certified deployment without per-case regulatory approval. ETSI GS ZSM 009 is mandated within EU NaaS federation governance frameworks. A CEN/CENELEC-standardised adversarial robustness test suite for security AI is published. MITRE ATT&CK behavioural framework is formally referenced in NIS2 sector-specific incident handling guidance. EuroHPC AI Factory compute contracts for EU-sovereign security model training are active. Without operational AI Act conformity assessment infrastructure, autonomous security agents cannot be certified for regulated-sector deployment regardless of technical readiness. This is the dominant Phase 2 policy dependency.	Certified EU sovereign AI agents in critical infrastructure SOC EU SLM-based LLMSecOps agents are in certified operational deployment in energy, financial (DORA), and telecommunications critical infrastructure SOC in ≥ 3 Member States. Guardian nodes execute autonomous Zero-Trust reconfigurations under human-in-the-loop escalation protocols for Tier-1 critical actions. NWDAF-integrated analytics are mandatory in 5G SA deployments. Deception infrastructure (honeynets, credential canaries, fake API endpoints per CL2.6) is deployed alongside AI detection systems, providing high-fidelity ground truth for model calibration. APT dwell time ≤ 72 hours demonstrated in pilot critical infrastructure deployments. Phase 2 deployment evidence requires named critical infrastructure sector, Member State scope, and a documented APT dwell time metric from an operational — not simulated — environment.	SSL: Sovereign EU Edge LLMs Emerging EU-sovereign SLMs cover $\geq 30\%$ of critical infrastructure SOC deployments in regulated sectors, rising from zero at baseline. EuroHPC AI Factories supply training infrastructure for EU-origin security models, reducing dependence on US hyperscaler compute. US-platform SIEMs retain market share in legacy deployments but face procurement competition from certified EU alternatives. Adversarial robustness validation is a commercially available service with EU-domiciled providers. EU AI security tool exports to partner jurisdictions have begun, indicating the transition from security consumer to security producer posture. SSL evidence at Phase 2: $\geq 30\%$ critical infra SOC market share for EU-sovereign AI tools; at least one documented EU AI security tool export event to a non-EU partner jurisdiction.

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SSL	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Sovereignty
	R&I trigger: EU-certified swarm AI coordination protocols enabling decentralised autonomous threat intelligence synthesis have passed independent security evaluation and are deployed in at least one cross-border EU critical infrastructure corridor under ETSI GS ZSM 009 governance. Policy trigger: EU AI security standards (adversarial robustness profiles, autonomous agent accountability frameworks) are adopted or formally referenced in at least one ISO/IEC or ITU-T international standard, confirming EU regulatory leadership beyond domestic jurisdiction. Operational trigger: APT dwell time ≤ 24 hours is documented in a named NIS2-essential entity operational deployment (not a simulation), attributable to autonomous AI-driven detection and containment. Market trigger: EU-sovereign AI security tools hold $\geq 50\%$ market share in new critical infrastructure SOC deployments across at least five Member States, displacing US-platform SIEMs as the procurement default in regulated sectors.			
al in y AI	Cluster ODP: Ubiquitous AI Immune System All three sub-dimensions reach Ubiquitous Operation. Swarm-based agentic AI defends the EU Cloud-Edge Continuum via self-evolving, decentralised threat intelligence without any central coordination dependency. Moving Target Defense continuously shuffles IP addressing, network topology, and software diversity; validated APT reconnaissance success rates fall below 5% in certified operational environments. Adversarial robustness validation is continuous and fully automated: no security AI agent operates in EU critical infrastructure without an active conformity certification. Phase 3 R&I addresses post-swarm frontiers: quantum-native AI security architectures, brain-inspired neuromorphic threat detection, and AI-to-AI adversarial dynamics beyond current threat models. <i>Phase 3 ODP evidence requires operational statistics from production deployments: APT dwell time figures, MTD effectiveness metrics, and autonomous containment accuracy rates from certified environments — not from cyber range simulations.</i>	EU AI security governance as international reference EU AI Act frameworks for trustworthy autonomous security agents are adopted internationally and referenced in bilateral digital partnership agreements. Autonomous agent certification operates at continental scale without per-case regulatory review. Zero-touch network management under ETSI GS ZSM 009 governs all EU NaaS operations. The EU AI Office and ENISA jointly define the international standard for security AI accountability. EU AI security governance has transitioned from reactive adaptation to agenda-setting: other jurisdictions adopt EU frameworks to access EU market and partnership programmes. <i>Phase 3 policy leadership is verified by international adoption events — not by the number of EU instruments in force. At least one non-EU jurisdiction formally adopting EU security AI governance standards constitutes the Phase 3 policy evidence threshold.</i>	Autonomous continental-scale AI cyber defence as operational default APT dwell time ≤ 24 hours in all certified EU NIS2-essential critical infrastructure deployments. MTD is deployed as a mandatory operational baseline. Swarm AI coordination operates without central cloud dependency: each EU Cloud-Edge node participates in a self-organising threat intelligence mesh. Deception infrastructure is fully integrated into the autonomous response loop, providing continuous model calibration ground truth. Human intervention is reserved for exceptional legal escalation scenarios; routine security operations are fully autonomous within the governance framework. <i>The Phase 3 deployment test: is autonomous AI defence the unconditional operational baseline, or is it still a premium programme? 100% of new critical infrastructure deployments incorporating autonomous AI security constitutes the evidence threshold.</i>	SSL: Global Leadership in Trustworthy AI Security EU-sovereign AI security tools hold $\geq 50\%$ of new critical infrastructure SOC deployments across the EU. US-platform SIEMs are legacy infrastructure under active replacement programmes with defined timelines. EU AI security models are exported to and commercially adopted by partner jurisdictions. EU sets the commercial terms for security AI market access: certification against EU AI Act Article 9/15 profiles is a de facto global procurement requirement. The EU has transitioned from defensive sovereignty to offensive strategic posture: it defines the technical terms on which the global security AI industry must compete. <i>Full sovereign leadership in this cluster requires both domestic dominance ($\geq 50\%$ critical infra SOC market share) and international influence (at least two partner jurisdictions formally adopting EU security AI governance). Both conditions must be met.</i>

10 Priority Cluster 3: Federated Operational Sovereignty

Constituent Sub-Priorities: P4 (Security Interoperability via Simpl Middleware), P5 (Supply Chain Verification and Secured Infrastructure), P7 (Pan-European NaaS Federation — “Schengen for Cloud-Edge/AI”).

Cluster Thesis: Technical excellence in Cluster 1 and autonomous defence capability in Cluster 2 deliver no strategic value if they cannot operate verifiably across borders, across providers, and across supply chains. This cluster defines the governance fabric, interoperability standards, and supply chain integrity mechanisms that transform individually sovereign components into a collectively sovereign European computing continuum. Critically, it addresses the legal and semantic as well as the technical interoperability problem — without a “Schengen for Cloud” legal framework, cross-border compute federation remains commercially fragile regardless of technical standardisation.

7-Dimensional Analysis: Priority Cluster 3

1. Baseline & Target Objective

Baseline (2025–2026): European cloud and edge infrastructure remains highly fragmented across proprietary provider silos with incompatible security policies, authentication mechanisms, and data exchange protocols. Common European Data Spaces (Manufacturing, Health, Energy, Agriculture) are in early piloting stages, with security interoperability relying on bilateral agreements rather than standardised automated mechanisms aligned with the Simpl open-source middleware stack [42]. Supply chain verification is largely static: CRA-mandated Software Bills of Materials (SBOMs) [8] are point-in-time declarations without runtime verification; Digital Product Passport (DPP) frameworks remain in pilot stage under the EU Ecodesign Regulation. NaaS federation across EU telecom operators does not exist at operational scale — cross-border edge compute sharing requires bespoke commercial agreements. CAMARA Network APIs are being standardised by GSMA but remain fragmentedly implemented across European operators.

Current cluster ODP: Fragmented Piloting (P4, P7); early supply chain disclosure phase (P5). Cluster SSL: Emerging EU Ecosystem (P4); Hyperscaler Platform Dominance (P7); High reliance on fragmented non-EU supply chain auditing (P5).

Target (2040): A unified European NaaS platform enables seamless cross-border compute roaming under a “Schengen for Cloud” legal framework, legally operable as a single jurisdiction for data workloads. All Common European Data Spaces operate on Simpl-based security interoperability with automated MIM compliance — provider-agnostic, cryptographically verifiable, and free of vendor lock-in. Runtime SBOMs with Zero-Knowledge Proof (ZKP) verification eliminate unverified components from EU critical infrastructure supply chains. EU achieves full sovereign control over supply chain transparency with pan-European Cyber Threat Intelligence sharing for supply chain compromise indicators.

2. Primary Drivers

Threat Drivers:

- Supply chain attacks exploiting the disaggregation complexity of Open RAN deployments, where a single compromised firmware component sourced from one of 50-plus

hardware and software vendors per macro cell deployment can affect thousands of base stations across multiple operators before detection.

- “Weakest link” cascading failures in federated NaaS environments: a single provider with a compromised Minimal Interoperability Mechanism (MIM) implementation can propagate malicious policies across the entire federated fabric before semantic validation detects the enforcement inconsistency.
- US CLOUD Act extra-territorial reach over non-EU cloud providers processing EU data creates a structural sovereignty risk that persists regardless of technical security measures if the legal framework enabling EU-sovereign federated compute is not established.
- Silent policy translation failures: automated security policy translation between heterogeneous provider MIM implementations may produce semantically equivalent but enforcement-different policies, creating exploitable gaps invisible to standard compliance auditing procedures.

Regulatory Drivers:

- Data Act interoperability obligations (Articles 23–31): mandatory requirements for cloud service providers covering security interface standardisation and data portability [7]. [CITATION REQUIRES VERIFICATION — confirm final Data Act article numbering for interoperability provisions.]
- NIS2 Article 21(2)(h): supply chain security measures mandatory for essential entities, requiring documented risk assessment of all ICT suppliers [7].
- CRA Annex I Part II: supply chain vulnerability handling requirements for products with digital elements, including SBOM disclosure obligations [8].
- DORA Article 28: ICT third-party risk management requiring continuous monitoring and documented exit strategies for critical ICT suppliers [44].

3. High Impact Activities

- (a) **[Phase 1]** Define and standardise security-specific Minimal Interoperability Mechanisms (MIMs) within the Simpl open-source middleware stack [42]; deploy pilots in three Common European Data Spaces (Manufacturing, Health, Energy) with automated cross-provider policy enforcement, federated identity attestation, and cryptographically verifiable compliance reporting.
- (b) **[Phase 1–2]** Implement Runtime SBOMs with cryptographic proof of execution for all containerised microservices and AI components migrating across the multi-provider continuum; integrate Digital Product Passports (DPP) for hardware component traceability from silicon fabrication to operational deployment, aligned with CRA SBOM obligations [8].
- (c) **[Phase 1–2]** Deploy harmonised CAMARA Network APIs across a minimum of five EU telecom operators establishing standardised interfaces for cross-border edge compute roaming; test Automated Compliance Negotiators in designated EU 5G cross-border corridor deployments under IPCEI-CIS governance.

- (d) **[Phase 2]** Establish “Data Visa” mechanisms enabling dynamic cross-border AI model and data migration without manual legal friction, governed by a common semantic security ontology (CL3.4) aligned with STIX 2.1/TAXII threat intelligence standards and automated EUCS compliance verification [10].
- (e) **[Phase 2–3]** Deploy Zero-Knowledge Proofs (ZKPs) for IP-protected supply chain component verification: hardware vendors prove FIPS 203/204/205 algorithm compliance and absence of known vulnerabilities without revealing proprietary design details [48–50]; operationalise pan-European supply chain compromise indicator sharing via standardised CTI lifecycle platforms (CL2.5).

4. Catalysts & Enablers

Funding Alignment: IPCEI-CIS (Important Project of Common European Interest on Next Generation Cloud Infrastructure and Services) as the primary funding vehicle for cross-border NaaS federation pilots; Digital Europe Programme for Simpl development, Common Data Space pilots, and MIM standardisation infrastructure; ECCC Strategic Agenda priority areas “Supply Chain Security” and “Secure Interoperability of Systems and Networks” [41]; European Alliance for Industrial Data, Edge and Cloud for governance and pilot ecosystem coordination [42].

Standardisation: CEN/CENELEC JTC 13 for security MIM standardisation within the Simpl ecosystem; GSMA CAMARA API Alliance for harmonised Network API standardisation across operators; ETSI TC CYBER for supply chain security standards; OASIS STIX 2.1/TAXII for threat intelligence interoperability; W3C Verifiable Credentials Data Model for SSI-based supply chain attestation chains.

Ecosystem Enablers: Gaia-X Federation Services trust framework alignment with Simpl MIM security policies; Digital Product Passport pilot programmes under EU Ecodesign Regulation 2024; ENISA supply chain risk management guidelines and threat landscape reports for ICT sectors; dedicated SME compliance support programmes for CRA-mandated SBOM tooling, recognising that sub-50 person software vendors represent the most vulnerable link in the supply chain verification architecture [8].

5. Disruptors & Systemic Risks

- **Hyperscaler resistance to Simpl MIM adoption:** AWS, Azure, and GCP collectively control approximately 65% of the EU cloud market as of 2024 and have strong commercial incentives to maintain proprietary APIs; without mandatory EU Cloud Rulebook provisions requiring MIM compliance, voluntary adoption rates are unlikely to achieve genuine interoperability within Phase 1 timelines.
- **IPCEI-CIS implementation delays:** complex EU state aid notification procedures have historically delayed IPCEI projects by 18–36 months (IPCEI-H2 delayed by state aid clearance processes); P7 NaaS federation Phase 1 milestones carry material schedule risk if IPCEI-CIS governance does not streamline implementation mechanisms.
- **Semantic interoperability gap:** automated security policy translation between heterogeneous MIM implementations is architecturally harder than syntactic interoperability — subtle semantic differences in “deny” versus “drop” versus “quarantine” semantics across provider implementations can create silent enforcement gaps undetectable through standard compliance auditing.

- **ZKP computational overhead:** current ZKP schemes for supply chain verification carry 10–100× computational overhead relative to standard cryptographic verification, limiting practical deployment to batch verification rather than real-time per-transaction attestation in Phases 1–2.
- **EU semiconductor supply chain concentration:** advanced node fabrication (sub-7 nm) required for cutting-edge supply chain attestation hardware remains concentrated in Taiwan (TSMC) and South Korea (Samsung); a Taiwan Strait contingency would immediately constrain EU supply chain verification capability at the silicon level, exposing a gap the EU Chips Act does not address within the 2030 timeframe.

6. Failsafe Mechanisms & Resilience

- **Autonomous unfederation:** if a provider fails Runtime SBOM verification or MIM compliance checks, eBPF-based zero-trust policy enforcement severs that provider from the federated NaaS fabric and redirects workloads to verified compliant nodes within a sub-60-second SLA; provider re-admission requires fresh ZKP-verified attestation, not manual recertification.
- **Graceful national fallback:** cross-border SLA failure or “Data Visa” mechanism collapse triggers workload redirect to national infrastructure maintained at sufficient sovereign capacity for essential services, fulfilling NIS2 Article 21(2)(c) business continuity obligations and DORA Article 9 ICT resilience requirements [7, 44].
- **Cryptographic attestation of MIM enforcement:** all Simpl MIM security controls include verifiable proofs that policies are enforced as declared — not merely configured — using TEE-backed attestation integrated with Cluster 1 secure element infrastructure [42], eliminating compliance-theatre scenarios where declared and actual enforcement diverge.
- **Supply chain quarantine:** unverified or high-risk components are isolated into quarantined network slices (VLAN/VxLAN-isolated segments with eBPF-enforced east-west traffic blocking) pending ZKP verification; quarantined components cannot interact with production workloads or access sensitive data paths.
- **Multi-jurisdictional residency fallback:** if cross-border “Data Visa” mechanisms fail legally or technically, automated GDPR Article 44 compliance enforcement pins workloads to the originating jurisdiction, preventing unauthorised cross-border data transfers under federation failure conditions [51].

7. Transversal Meta-Layer

Sovereignty & Autonomy: This cluster breaks non-EU hyperscaler API lock-in (AWS, Azure, GCP controlling approximately 65% of EU cloud market as of 2024) by enabling European providers to compete on standardised Simpl-based infrastructure. A “Schengen for Cloud” legal framework eliminates US CLOUD Act extra-territorial jurisdiction exposure for EU data processed on federated EU-sovereign NaaS infrastructure. Runtime SBOM verification and ZKP-based supply chain attestation reduce EU dependency on non-EU vendor self-certification declarations — the dominant model under current CRA-baseline SBOM obligations.

Trust & Ethics: GDPR Article 25 (privacy by design and by default) is operationalised through MIM-enforced data minimisation policies that are cryptographically verifiable, not declarative [51]. Data Act interoperability obligations are fulfilled through Simpl MIM compliance, providing auditable trust chains for GDPR Article 5(2) data processing accountability. Transparent Runtime SBOM verification provides the supply chain integrity that GDPR Article 25 implicitly requires for processors handling EU personal data at scale [51].

Sustainable Security (Green IT): Federated NaaS resource pooling across EU operators eliminates redundant edge compute over-provisioning by individual operators, reducing total EU edge infrastructure carbon footprint through shared utilisation efficiency. DPP-based supply chain traceability aligns with EU Ecodesign Regulation circular economy requirements for ICT hardware, enabling component reuse lifecycle tracking that reduces electronic waste from premature infrastructure replacement driven by opacity in component provenance.

10.1 Progression Matrix: Priority Cluster 3 — Federated Operational Sovereignty

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SSL	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Sovereignty
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Table 5: **Cluster 3: Federated Operational Sovereignty** (P4, P5, P7). Transition matrix across the four methodology dimensions. P4 (Simpl Interoperability), P5 (Supply Chain Verification), and P7 (NaaS Federation) are labelled where their maturity trajectories diverge within a cell.

SSL	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Sovereignty
	ODP indicator per sub-priority	enforcement · certification · standardisation	actor types · scale · sector coverage	SSL indicator · dependency reduction · industrial capacity
Cluster 1 (P4, P5, P7)	Cluster ODP: Fragmented Piloting (P4, P7) / Disclosure Phase (P5) (P4) is at Fragmented Piloting: Simpl security MIM pilots are launched in three Common European Data Spaces (Manufacturing, Health, Energy) with manual policy coordination and proof-of-concept implementations. (P7) is at Bilateral Interconnection: harmonised CAMARA Network APIs are deployed in 2–3 bilateral 5G corridor agreements between individual EU telecom operators; cross-border edge compute roaming does not exist at operational scale. (P5) is at Supply Chain Disclosure: CRA-mandated static SBOM declarations are being submitted as point-in-time compliance artefacts without runtime verification; Digital Product Passport pilot programmes are initiated under the EU Ecodesign Regulation; Zero-Knowledge Proof verification for supply chain attestation is at Applied R&D stage. The three sub-priorities have meaningfully different Phase 1 maturity levels. P5 is the least mature and the most consequential shared dependency: Runtime SBOM verification is a gate condition for both P4 (MIM compliance trust) and Cluster 1 (TEE integrity claims).	Multiple legislative obligations activating; standards nascent CRA SBOM obligations (Annex I Part II) enter enforcement in 2027, mandating static SBOM disclosure for all products with digital elements entering the EU market. NIS2 Article 21(2)(h) makes supply chain risk assessment mandatory for essential entities. DORA Article 28 requires continuous ICT third-party risk monitoring for financial sector entities. Data Act interoperability obligations activate, creating procurement-level pressure for cross-provider security interface standardisation. The Simpl-Open official launch with initial MIM definitions constitutes the Phase 1 technical policy trigger for P4. IPCEI-CIS enters its deployment phase. GSMA CAMARA API Alliance standardisation work is active but not yet mandated in EU regulatory instruments. Phase 1 policy activity is high-volume but fragmented: multiple legislative instruments activate simultaneously without a unified interoperability mandate that would create coherent market pull for Simpl-based standardisation.	Deployment: bilateral pilots and Data Space proof-of-concepts Three Common European Data Space security pilots operate Simpl-based MIM proof-of-concepts with manual policy coordination; cross-provider security policy enforcement is not automated. CAMARA API deployments are confined to 2–3 bilateral 5G corridor agreements between single operator pairs. Static SBOM declarations are submitted as CRA compliance artefacts but are not verified at runtime; supply chain risk assessment is vendor-declared rather than independently attested. Hyperscalers (AWS, Azure, GCP) retain ≈65% of EU cloud market through proprietary API ecosystems. Cross-border AI model migration requires bespoke bilateral legal review per transfer event. SME-accessible SBOM tooling and MIM compliance support are not yet standardised or commercially affordable. Phase 1 deployment is bilateral and pilot-scale. The gap between proof-of-concept MIM implementations and automated enforcement across the majority of Data Spaces defines the core operational challenge for Phase 2.	SSL: Mixed — Emerging EU Ecosystem (P4) / Hyperscaler Dominance (P7) (P4) has reached Emerging EU Ecosystem: the Simpl open-source stack exists and is being piloted with EU institutional backing. (P7) remains at Hyperscaler Platform Dominance: AWS, Azure, and GCP collectively control ≈65% of EU cloud market; no EU NaaS federation operates at commercially relevant scale; cross-border edge compute is fragmented into national silos. (P5) is at High Reliance on Fragmented Non-EU Auditing: supply chain verification is vendor-declared under US-dominated certification ecosystems (Common Criteria national schemes, SOC2). The cluster's dominant SSL profile is foreign dependency, driven by the weight of P7 market structure. P4 and P7 require separate SSL tracking in Phase 1: conflating them into a single cluster SSL figure obscures that Simpl has achieved Emerging Ecosystem while NaaS federation has not.

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SL	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Sovereignty
	R&I trigger: At least one Common European Data Space has deployed automated (not manually coordinated) Simpl MIM-based cross-provider security policy enforcement, with documented evidence that enforcement is cryptographically verifiable rather than declaratively asserted. Policy trigger: Simpl security MIM definitions are published in candidate CEN/CENELEC standard form and referenced as a compliance pathway in at least one EU sector-specific NIS2 or Data Act implementing act. Operational trigger: CAMARA-based cross-border edge compute roaming is operational between at least three distinct EU telecom operator pairs (not a single bilateral agreement), with documented SLA performance and compliance evidence. Market trigger: Runtime SBOM tooling with cryptographic execution proofs is available as a commercially supported product or service from at least two EU-domiciled vendors, establishing the market infrastructure for Phase 2 mandatory adoption.			

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SL	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Sovereignty
Industrial side gation ing y	Cluster ODP: Standardised Integration (P4) / Federated Edge Roaming (P7) / Dynamic Runtime Verification (P5) (P4) reaches Standardised Integration: automated policy translation via standardised Simpl MIMs is deployed across the majority of active Common European Data Spaces; federated identity attestation and cross-provider compliance reporting are automated. (P7) reaches Federated Edge Roaming: CAMARA-based cross-border roaming operates across ≥ 5 EU operator pairs; “Data Visa” mechanisms enable dynamic AI model migration without manual legal review in certified corridors. (P5) transitions from disclosure to dynamic runtime verification: Runtime SBOMs with cryptographic proof-of-execution verify containerised components as they migrate across multi-provider nodes; ZKP batch verification of supply chain attestation is operationally validated; Digital Product Passports are mandatory under the EU Ecodesign Regulation. Phase 2 R&I activity includes industrial R&D for ZKP real-time verification (currently batch-only due to 10–100× overhead), semantic security ontology standardisation, and legal engineering of the “Schengen for Cloud” framework — not solely standardisation of existing prototypes.	Security MIMs as CEN/CENELEC standards: the Phase 2 gate The single Phase 2 policy gate is the adoption of Simpl security MIMs as ratified CEN/CENELEC standards, referenced as mandatory compliance requirements in the EU Cloud Rulebook. This transforms MIM adoption from voluntary best-practice to market-access obligation for providers serving EU-regulated sectors. DPP becomes mandatory under the EU Ecodesign Regulation 2024 for ICT hardware categories. Data Act full enforcement creates binding procurement-level interoperability obligations. DORA continuous ICT third-party monitoring is fully operationalised in the financial sector. IPCEI-CIS cross-border NaaS governance framework is operational. <i>The Phase 2 policy gate is the CEN/CENELEC ratification of security MIMs. Voluntary Simpl adoption without this standards event does not constitute Phase 2 policy maturity.</i>	Cross-sector rollout; SME participation; Runtime SBOM in production Simpl MIMs are deployed across the majority of active Common European Data Spaces with automated cross-provider enforcement. Runtime SBOMs with cryptographic execution proofs verify supply chain components in critical infrastructure deployments in at least five Member States. CAMARA cross-border roaming operates across ≥ 5 EU operator pairs with documented SLA performance. “Data Visa” mechanisms are operational in certified cross-border AI model migration corridors. SME-accessible Simpl MIM compliance tooling and Runtime SBOM services are commercially available from EU-domiciled providers. Unverified supply chain components are automatically quarantined in isolated network slices. <i>Phase 2 deployment must show geographic breadth (≥ 5 Member States), cross-sector presence (≥ 3 Data Spaces under automated MIM enforcement), and SME participation in at least one deployment context. Single-sector rollout does not constitute Phase 2 operational maturity.</i>	SSL: Growing Pan-EU Sovereignty EU Data Space infrastructure operating under Simpl achieves $\geq 30\%$ non-hyperscaler share in new regulated-sector cloud procurement. Runtime SBOM ecosystem is commercially mature with multiple competing EU-domiciled vendors. EU NaaS consortium (under IPCEI-CIS) operates cross-border compute at pilot commercial scale, demonstrating that EU providers can federate to compete with hyperscaler geographic coverage. DPP traceability reduces procurement risk for EU critical infrastructure buyers by making supply chain provenance verifiable. Foreign hyperscaler market share in EU regulated sectors declines for the first time from its $\approx 65\%$ Phase 1 baseline. <i>Phase 2 SSL evidence: $\geq 30\%$ non-hyperscaler share in new regulated procurement; first documented market share decline for non-EU hyperscalers in EU regulated cloud procurement.</i>

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SL	R&I Maturity	Policy, Regulatory & Standards	Operational Deployment & Adoption	Market Uptake & Strategic Sovereignty
	R&I trigger: ZKP verification for supply chain attestation has achieved real-time per-transaction performance (eliminating the current batch-only constraint) in at least one production critical infrastructure deployment, confirmed by independent performance benchmarking. Policy trigger: A “Schengen for Cloud” legal framework — or equivalent EU legal instrument resolving US CLOUD Act extra-territorial jurisdiction conflict for federated EU-sovereign NaaS workloads — has been enacted, providing the jurisdictional basis for single-market operation of federated compute. Operational trigger: The EU NaaS federation (under IPCEI-CIS or its successor) operates at commercial scale across ≥ 10 EU telecom operator members, offering cross-border edge compute services as a standard commercial product with documented customer deployments outside its founding consortium. Market trigger: EU supply chain governance frameworks (Runtime SBOM plus ZKP attestation) are adopted or formally referenced by at least two non-EU partner jurisdictions as part of digital trade or partnership agreements, confirming that EU governance leadership is internationally exportable.			
	Cluster ODP: Ubiquitous Operation across all sub-priorities All three sub-priorities reach Ubiquitous Operation. Simpl-based security is the universal standard for cross-provider Data Space operation in the EU; no alternative interoperability model holds market relevance in regulated sectors. ZKP-verified supply chain attestation operates at real-time transaction scale for all critical infrastructure component verification. The EU NaaS federation operates as a single commercial entity legally equivalent to a unified jurisdiction for data workloads. Phase 3 R&I pursues next-generation frontiers: post-ZKP cryptographic verification, semantic AI-to-AI policy negotiation, and legal engineering of the next-generation “Digital Single Market” framework beyond cloud-edge compute. <i>Phase 3 ODP evidence: Simpl MIM compliance rate in new regulated EU cloud procurement; ZKP verification throughput in production critical infrastructure deployments; EU NaaS federation revenue relative to hyperscaler EU revenue in regulated sectors.</i>	EU federated governance as international model The “Schengen for Cloud” legal framework is enacted and operational, resolving US CLOUD Act jurisdiction conflicts for federated EU-sovereign workloads and enabling single-market legal operation of cross-border compute. EU Cloud Rulebook is mandatory for all new cloud deployments in EU regulated sectors without exception. EU supply chain governance standards (Runtime SBOM plus ZKP) are referenced in international supply chain security standards (ISO/IEC) and in digital trade agreements with partner jurisdictions. The EU has transitioned from rule-taker to rule-setter in international cloud and supply chain governance. <i>Phase 3 governance leadership is verified by two events: enactment of the “Schengen for Cloud” framework (domestic) and formal adoption of EU supply chain governance standards by ≥ 2 non-EU partner jurisdictions (international).</i>	Provider-agnostic EU trust fabric as continental operational default A single provider-agnostic EU trust fabric operates continuously across all active Common European Data Spaces with real-time automated security orchestration and zero manual policy coordination. ZKP-verified supply chain attestation eliminates all unverified components from EU critical infrastructure deployments in regulated sectors. The EU NaaS federation is a standard commercial service operating at hyperscaler-comparable scale, with EU operators collectively competitive on cost and coverage. Security-by-Design is not an add-on: every new cloud-edge deployment in the EU incorporates Simpl MIM compliance, Runtime SBOM verification, and cross-border sovereignty assurance as architectural defaults. <i>Phase 3 deployment normalisation: Simpl MIM compliance in $\geq 90\%$ of new regulated EU cloud deployments; ZKP supply chain verification in 100% of new critical infrastructure component procurement; EU NaaS federation operating as a single commercial entity with documented non-founding-member customers.</i>	SSL: Full Federated Sovereign Autonomy Non-EU hyperscaler market share in EU regulated cloud procurement falls below 40% for new deployments (from $\approx 65\%$ at baseline), with a binding trajectory toward continued reduction. EU NaaS federation achieves hyperscaler-comparable geographic coverage and commercial viability across the EU. Supply chain governance internationally adopted: at least two partner jurisdictions have formally adopted EU Runtime SBOM plus ZKP attestation standards as domestic procurement requirements. The EU has removed the structural dependency that allowed US CLOUD Act jurisdiction to apply to EU-regulated data: federated sovereignty is legally guaranteed, not only technically implemented. <i>Full Federated Sovereign Autonomy requires three concurrent conditions: $< 40\%$ hyperscaler new-procurement share; “Schengen for Cloud” legally enacted; ≥ 2 partner jurisdictions formally adopting EU supply chain governance standards. Achieving any two without the third does not constitute this SSL level.</i>

11 Cross-Priority Bottleneck Assessment

Three shared structural dependencies determine whether individual sub-priority progressions can advance on their stated timelines, regardless of cluster-level investment. These bottlenecks are identified here because they constitute cross-cutting gate conditions that cannot be resolved by any single cluster in isolation.

Bottleneck 1 — EU Chips Act Sovereign Fabrication Capacity. The Phase 2 gate conditions for P1 (EUCC ‘High’ certified RISC-V secure elements), P2 (PQC-native RISC-V silicon), P3 (zero-energy secure hardware), and P8 (RISC-V Keystone TEE scaling) all converge on the same physical dependency: sovereign EU fabrication capacity for security-critical silicon at commercially viable yields. The Phase 2 critical enabler for P1 (“EUCC ‘High’ certification of first EU-manufactured RISC-V secure element”) and the Phase 3 enabler (“EU Chips Act sovereign EUV-capable fabrication achieving $\geq 60\%$ self-sufficiency”) constitute a single two-stage gate that controls the SSL trajectory of four of the five Cluster 1 sub-priorities simultaneously.

Bottleneck 2 — AI Act Conformity Assessment Infrastructure. The Phase 2 gate condition for Cluster 2 (“standardised AI Act conformity assessment framework for autonomous security agents”) is a prerequisite not only for P6 and P10 but also for any AI-driven capability within Cluster 1 (autonomous Island Mode swarm protocols) and Cluster 3 (Automated Compliance Negotiators in NaaS federation) that incorporates autonomous decision-making. The absence of EU AI Act conformity assessment bodies with sufficient capacity to certify high-risk security AI at scale represents a systemic bottleneck that, if unresolved by 2028–2030, will delay the Phase 2 SSL transition across all three clusters.

Bottleneck 3 — Runtime SBOM and Supply Chain Attestation Infrastructure. Runtime SBOM verification (P5) is a prerequisite for both the Cluster 1 TEE integrity claim (P8 confidential computing pipelines cannot be trusted without verified component provenance) and the Cluster 3 MIM compliance model (Simpl-based interoperability cannot provide cryptographic enforcement guarantees if component integrity is unverifiable). The Phase 1 critical enabler for Cluster 3 (Simpl-Open MIM definitions) and the Phase 1 ODP for P5 (full supply chain disclosure) must therefore proceed in coordination, not sequentially, with the Phase 1 applied research activities in Cluster 1. This interdependency must be explicitly reflected in Horizon Europe project call design to ensure that SBOM infrastructure investment is not siloed within supply chain security calls but is structurally integrated into hardware security and interoperability funding instruments.

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